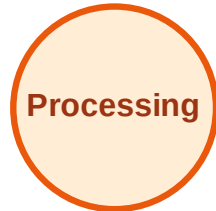


BFS Traversal

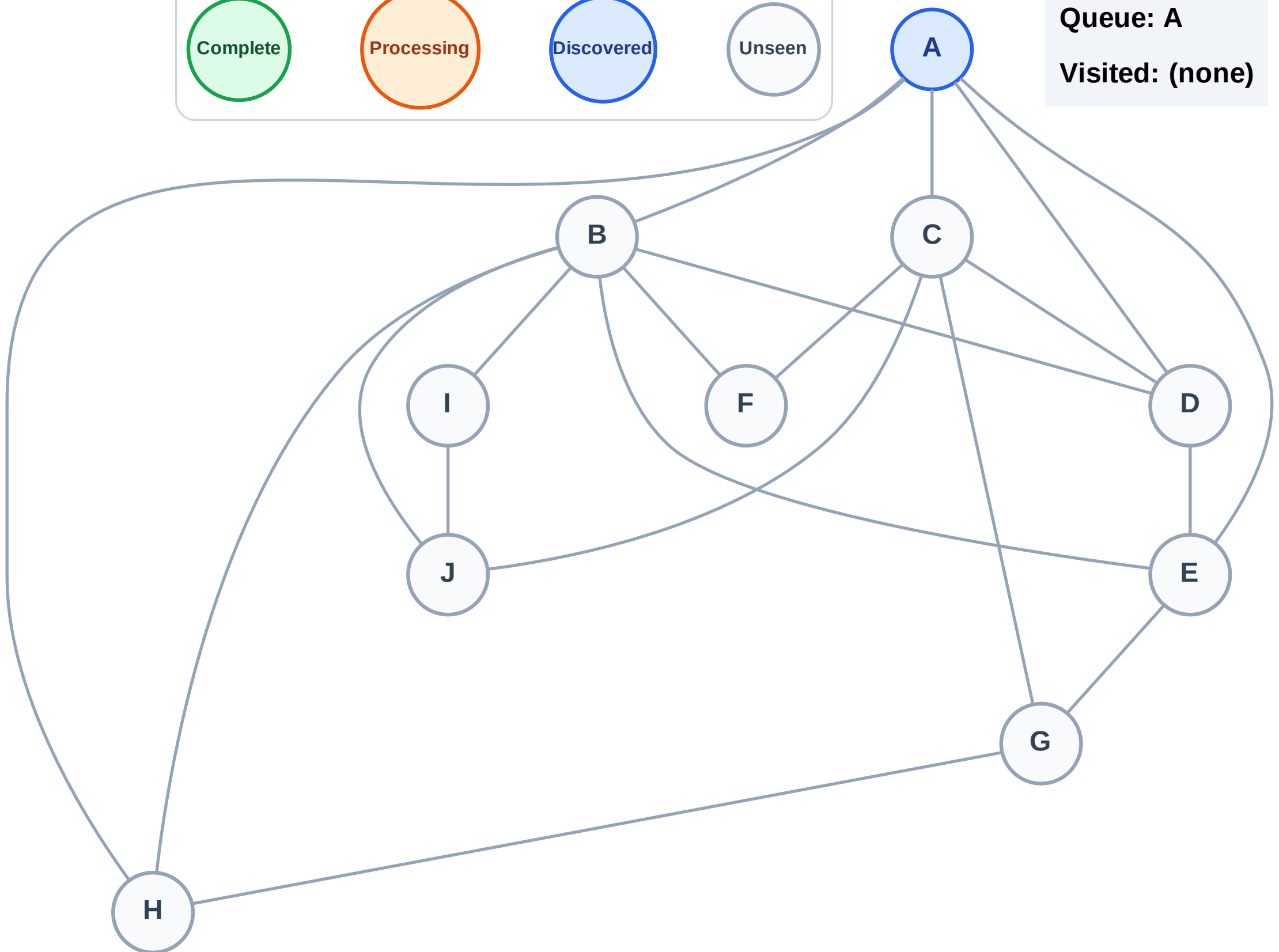
Step 1: Start at A

Legend



Queue: A

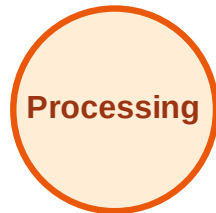
Visited: (none)



BFS Traversal

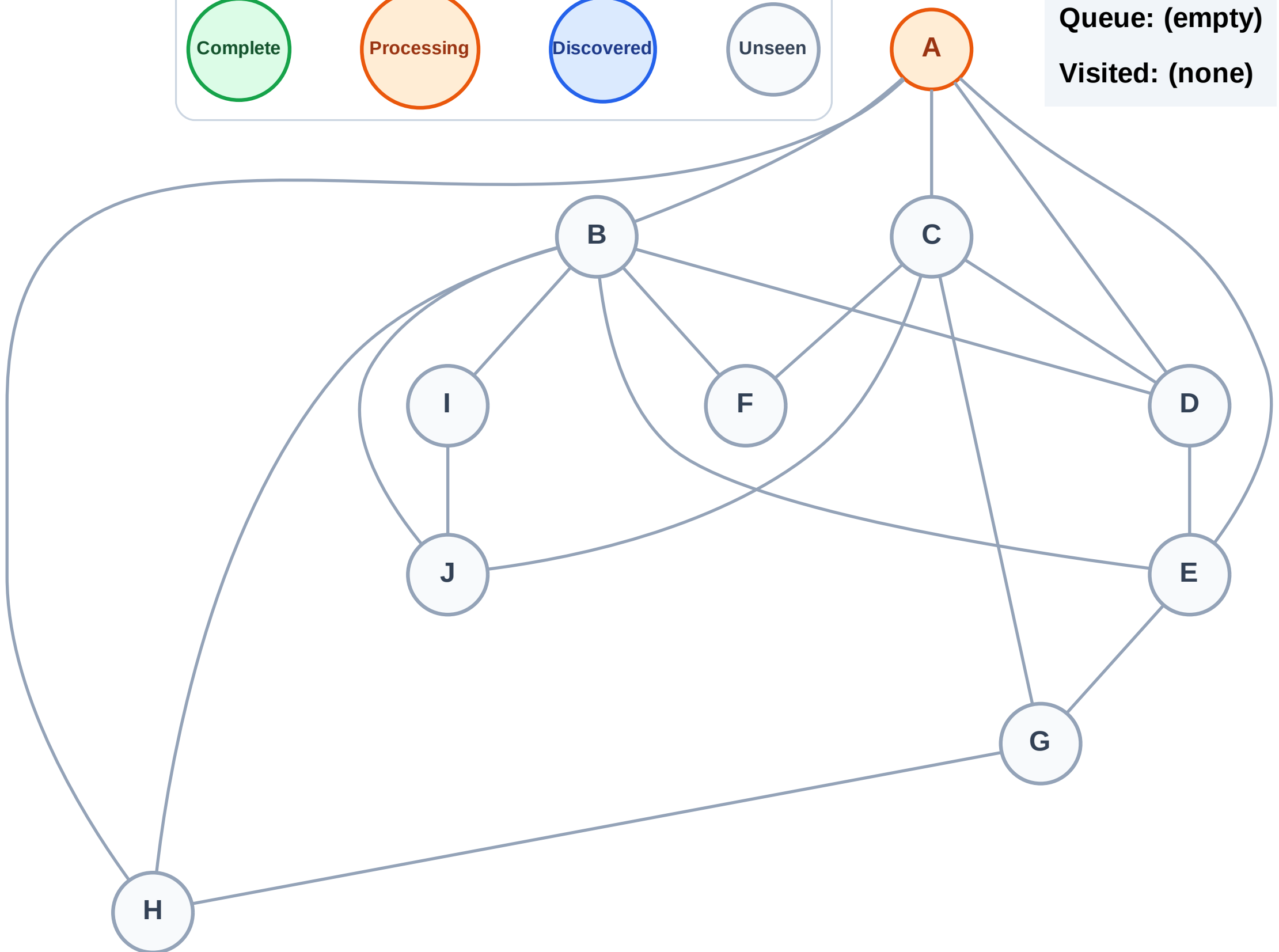
Step 2: Process node A

Legend



Queue: (empty)

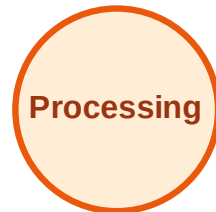
Visited: (none)



BFS Traversal

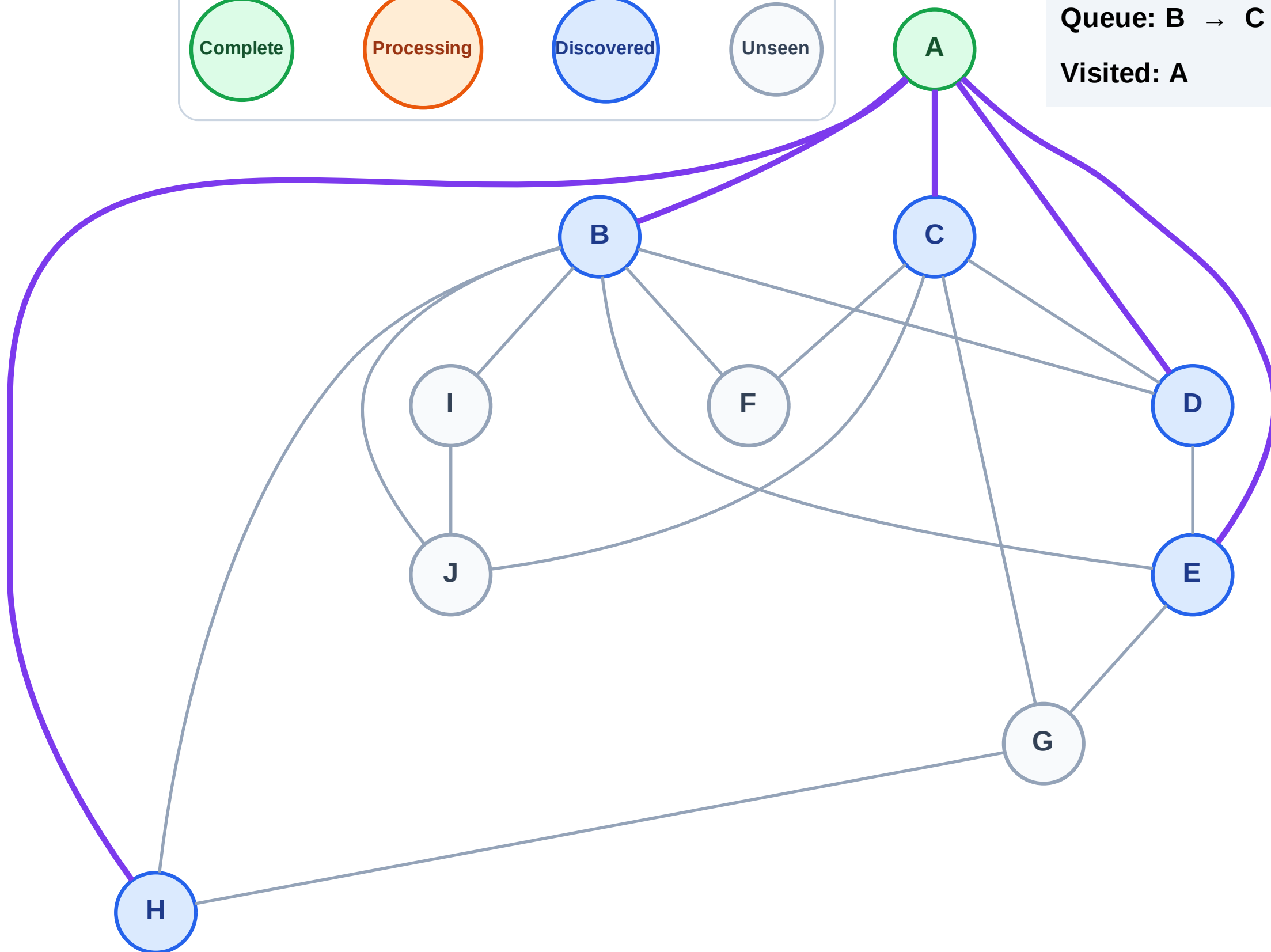
Step 3: Discover B, C, D, E, H from A

Legend



Queue: B → C → D → E → H

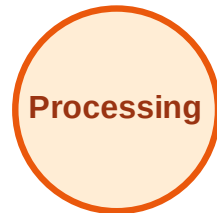
Visited: A



BFS Traversal

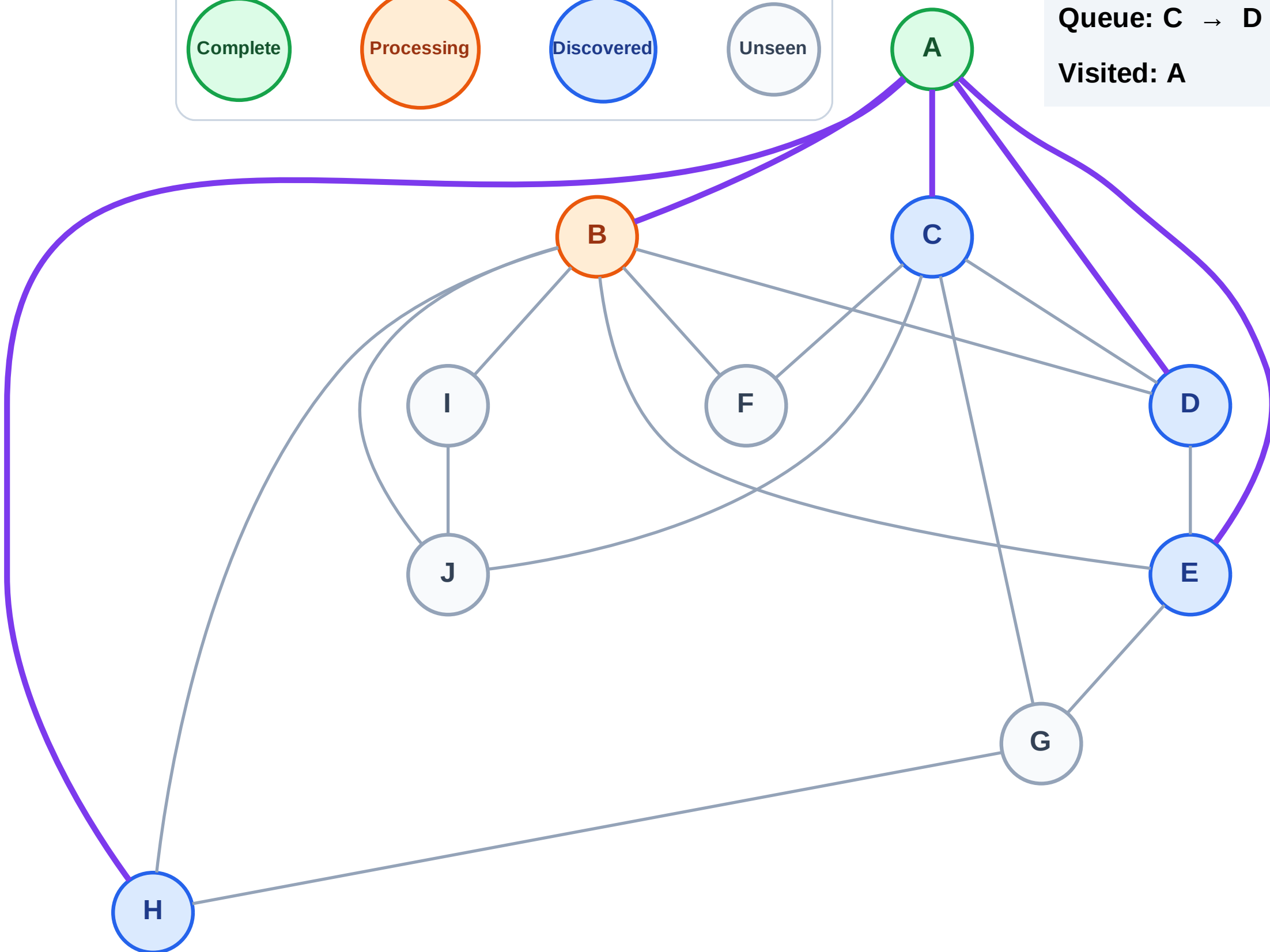
Step 4: Process node B

Legend



Queue: C → D → E → H

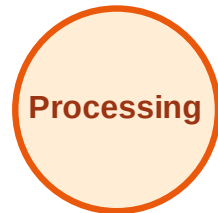
Visited: A



BFS Traversal

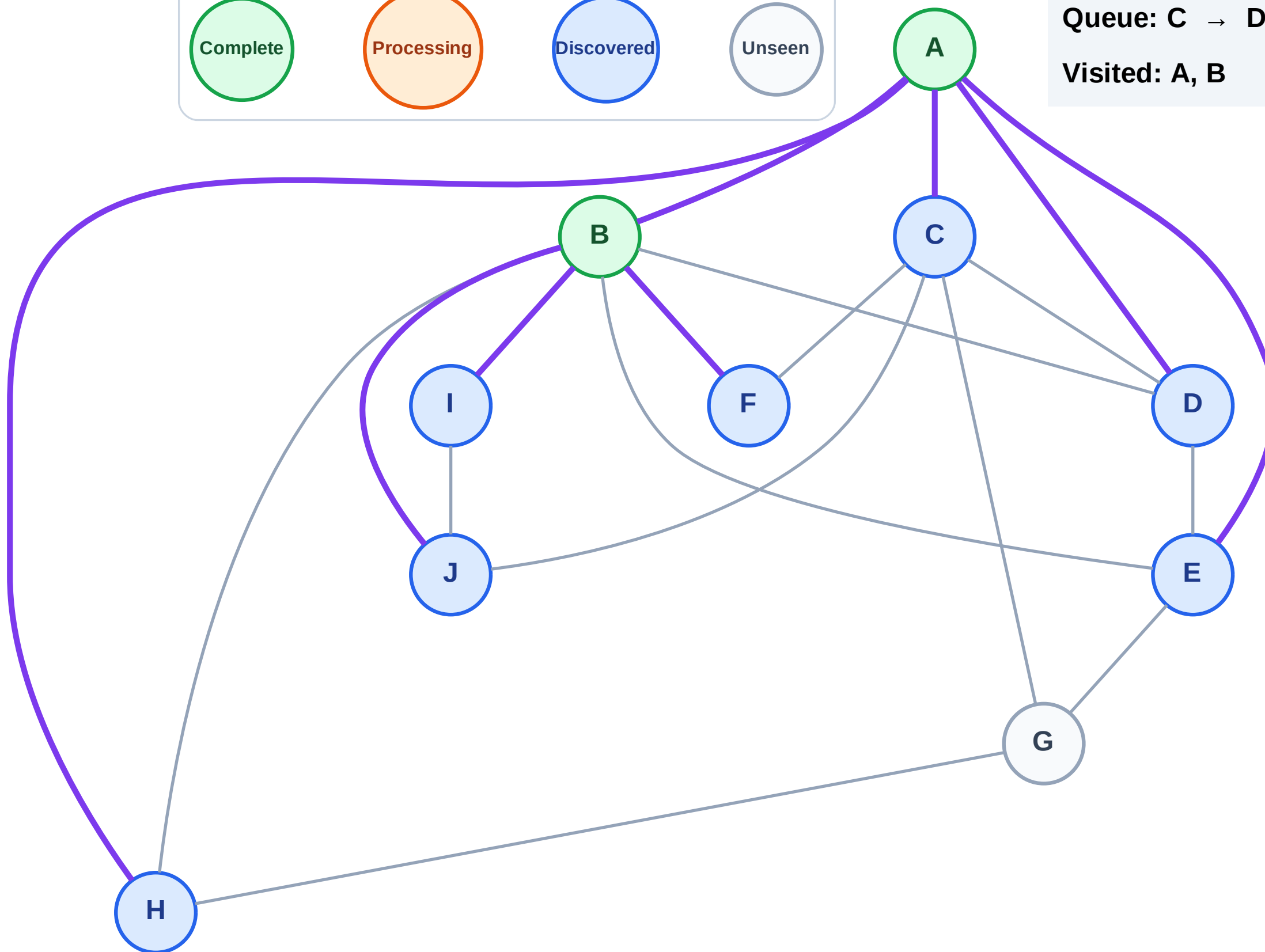
Step 5: Discover F, I, J from B

Legend



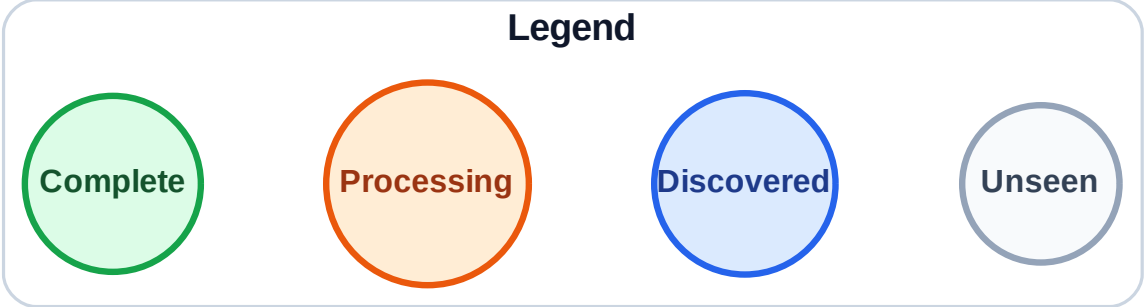
Queue: C → D → E → H → F → I → J

Visited: A, B



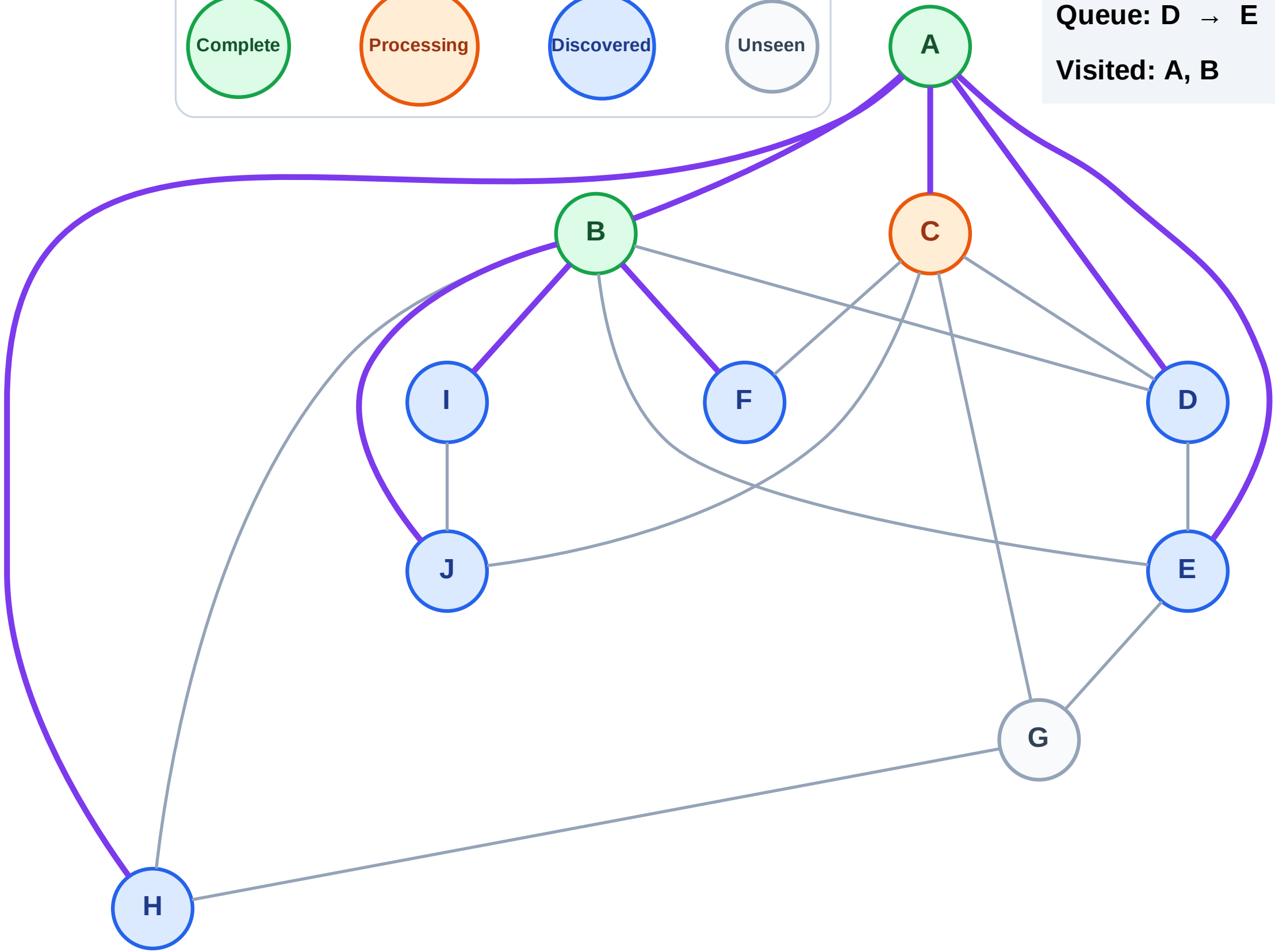
BFS Traversal

Step 6: Process node C



Queue: D → E → H → F → I → J

Visited: A, B



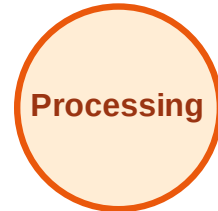
BFS Traversal

Step 7: Discover G from C

Legend



Complete



Processing



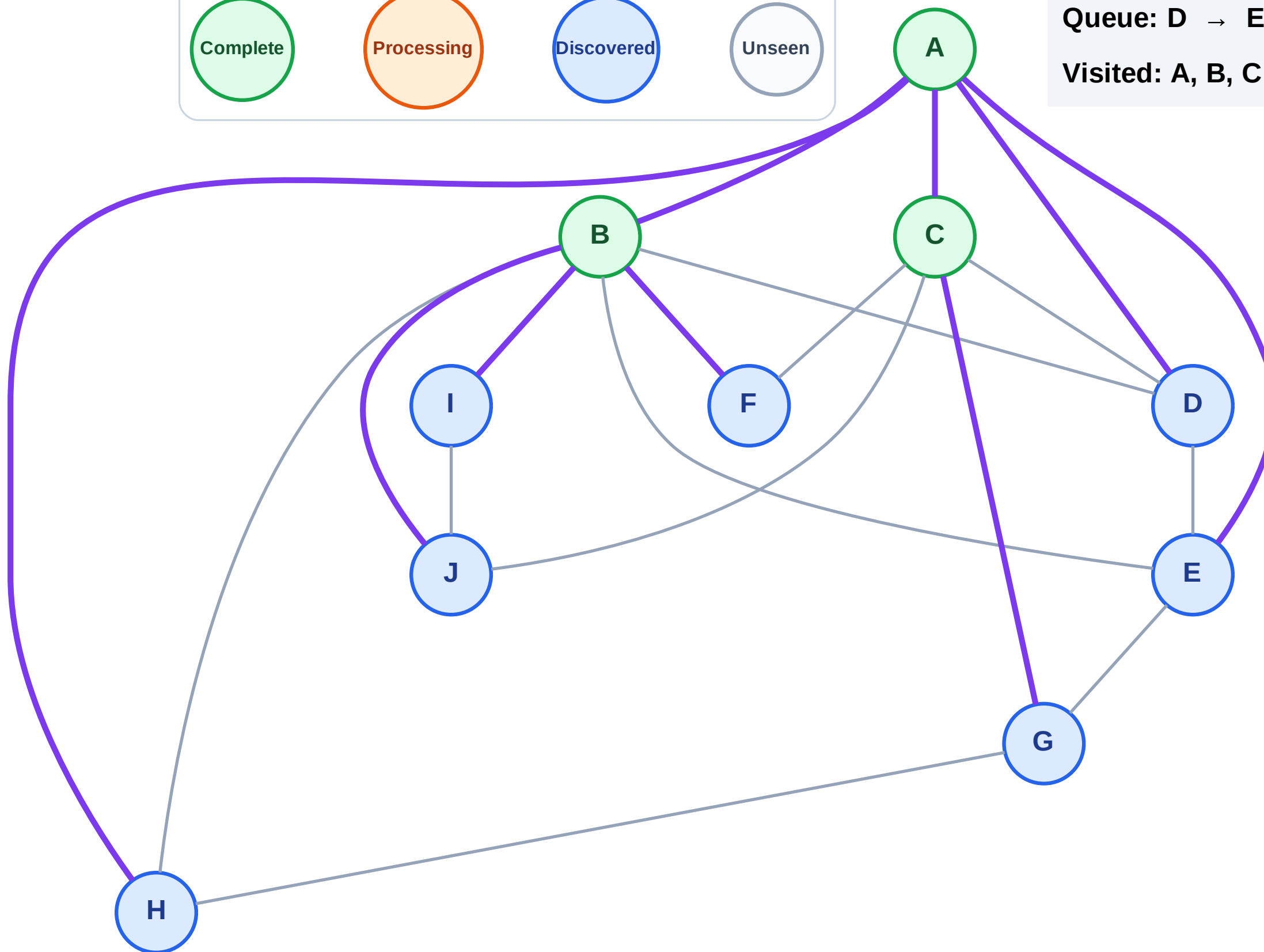
Discovered



Unseen

Queue: D → E → H → F → I → J → G

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

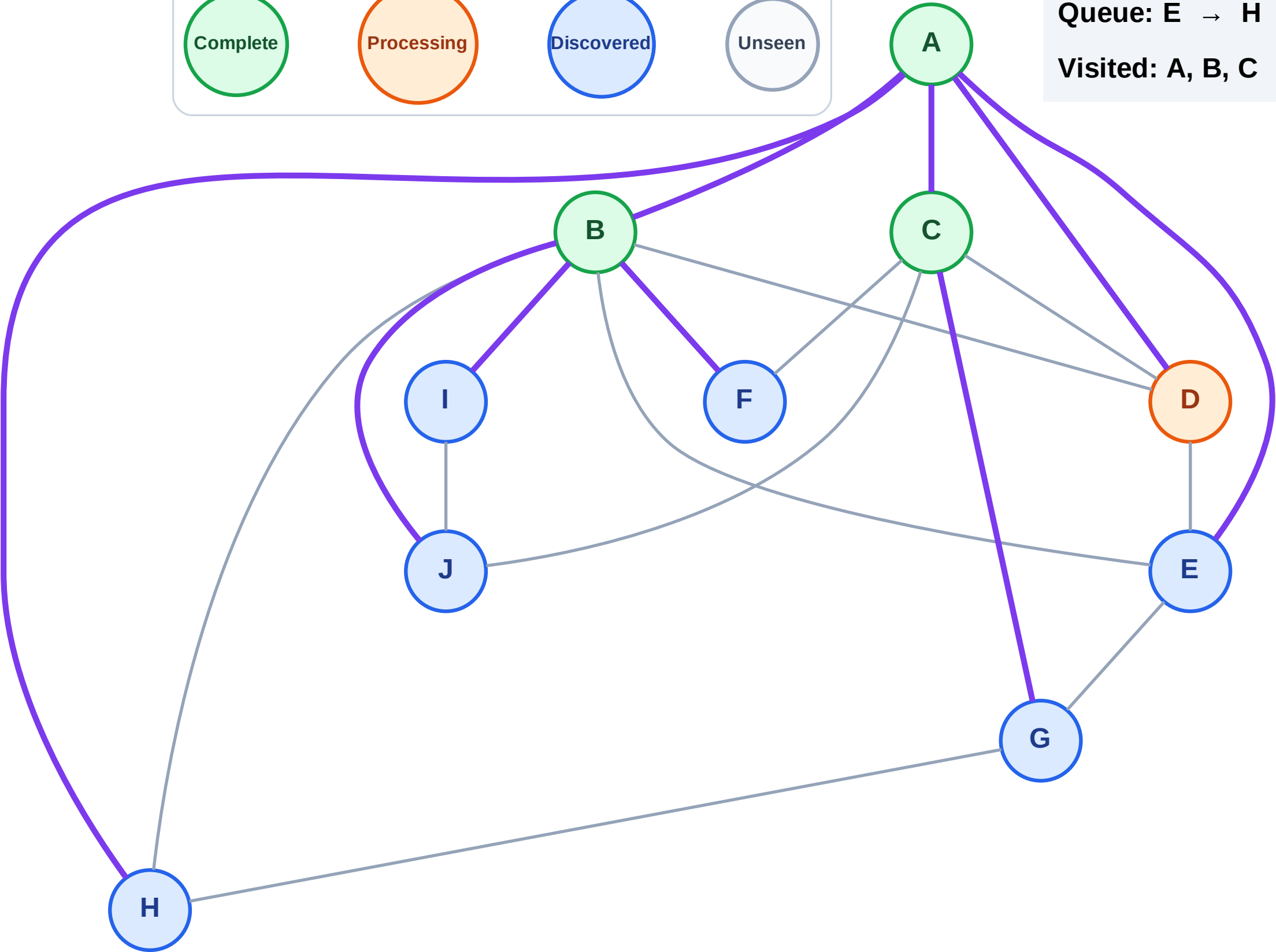
Processing

Discovered

Unseen

Queue: E → H → F → I → J → G

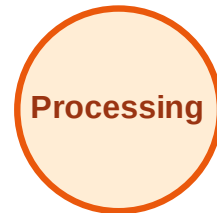
Visited: A, B, C



BFS Traversal

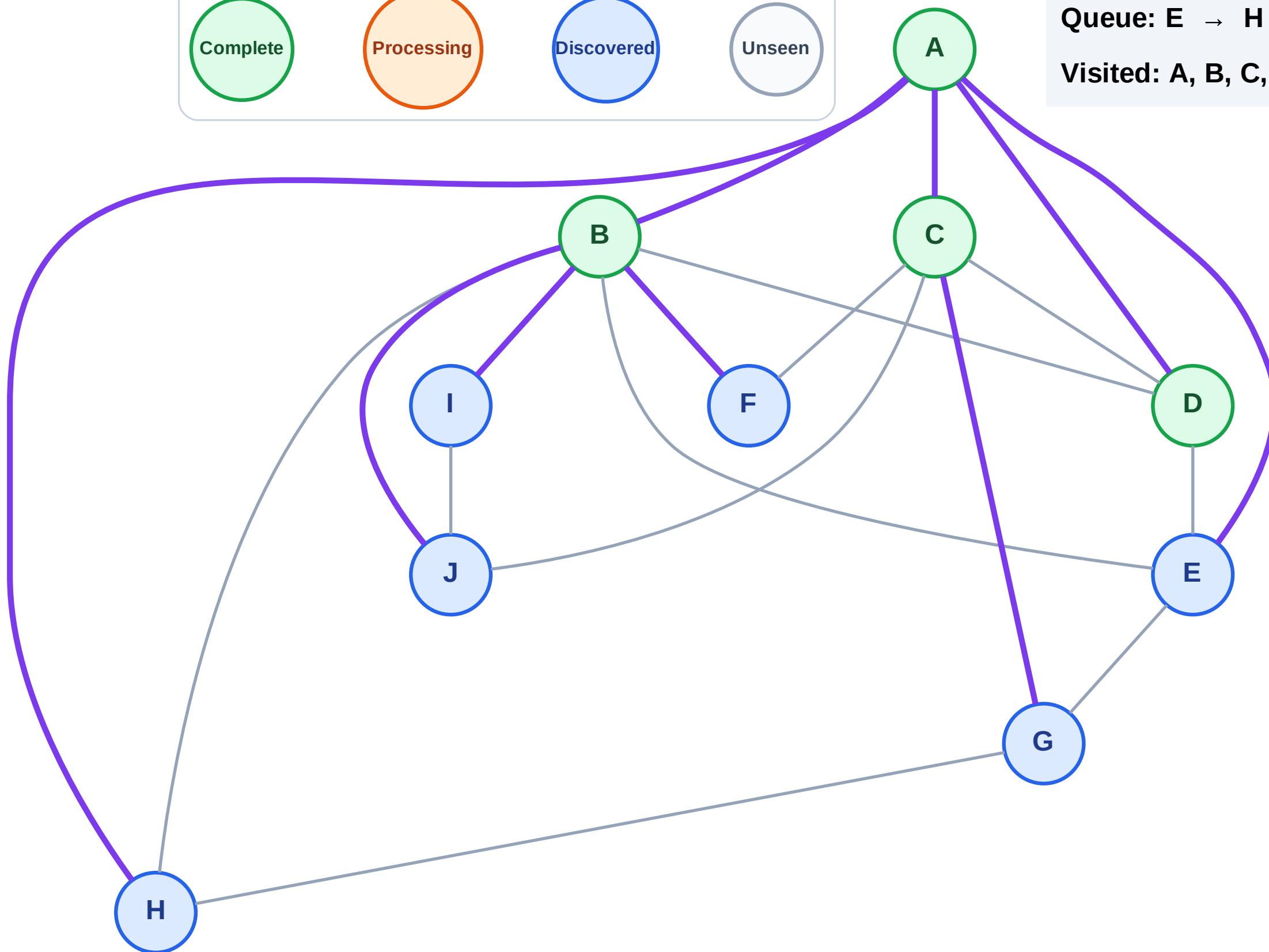
Step 9: No new neighbors from D

Legend



Queue: E → H → F → I → J → G

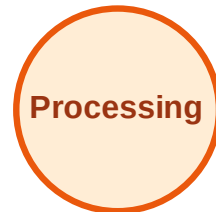
Visited: A, B, C, D



BFS Traversal

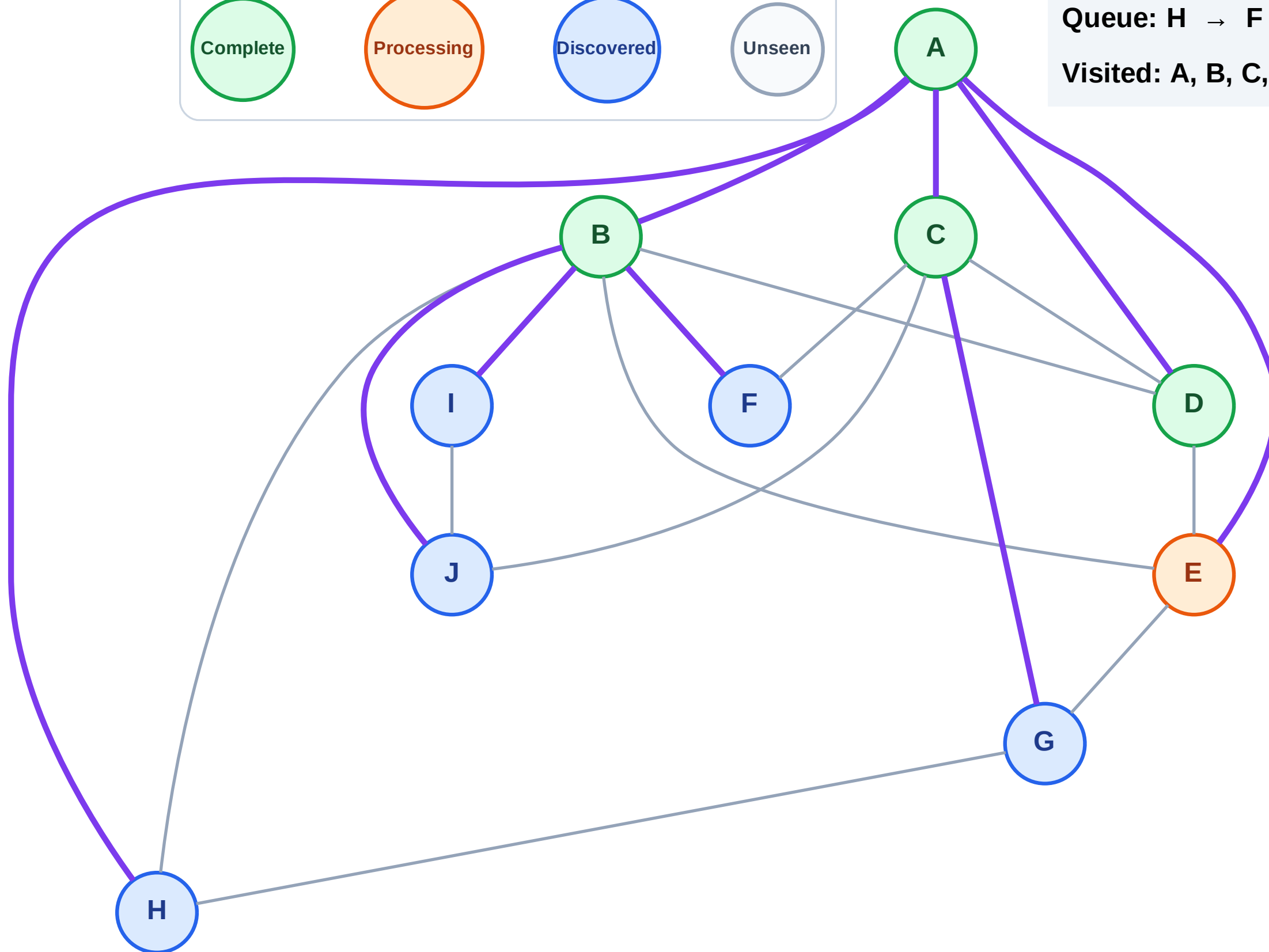
Step 10: Process node E

Legend



Queue: H → F → I → J → G

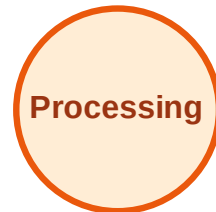
Visited: A, B, C, D



BFS Traversal

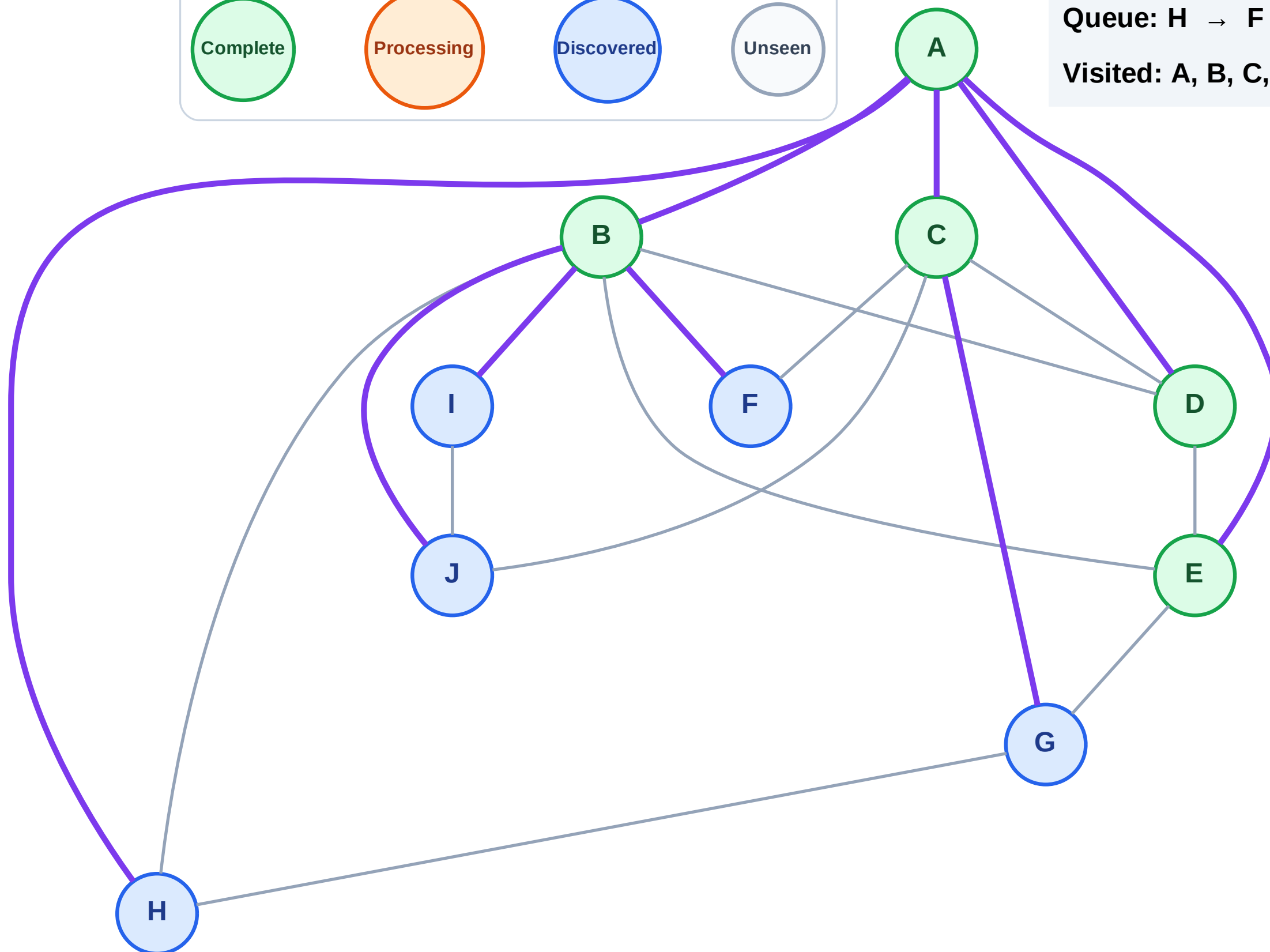
Step 11: No new neighbors from E

Legend



Queue: H → F → I → J → G

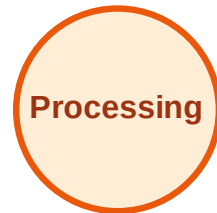
Visited: A, B, C, D, E



BFS Traversal

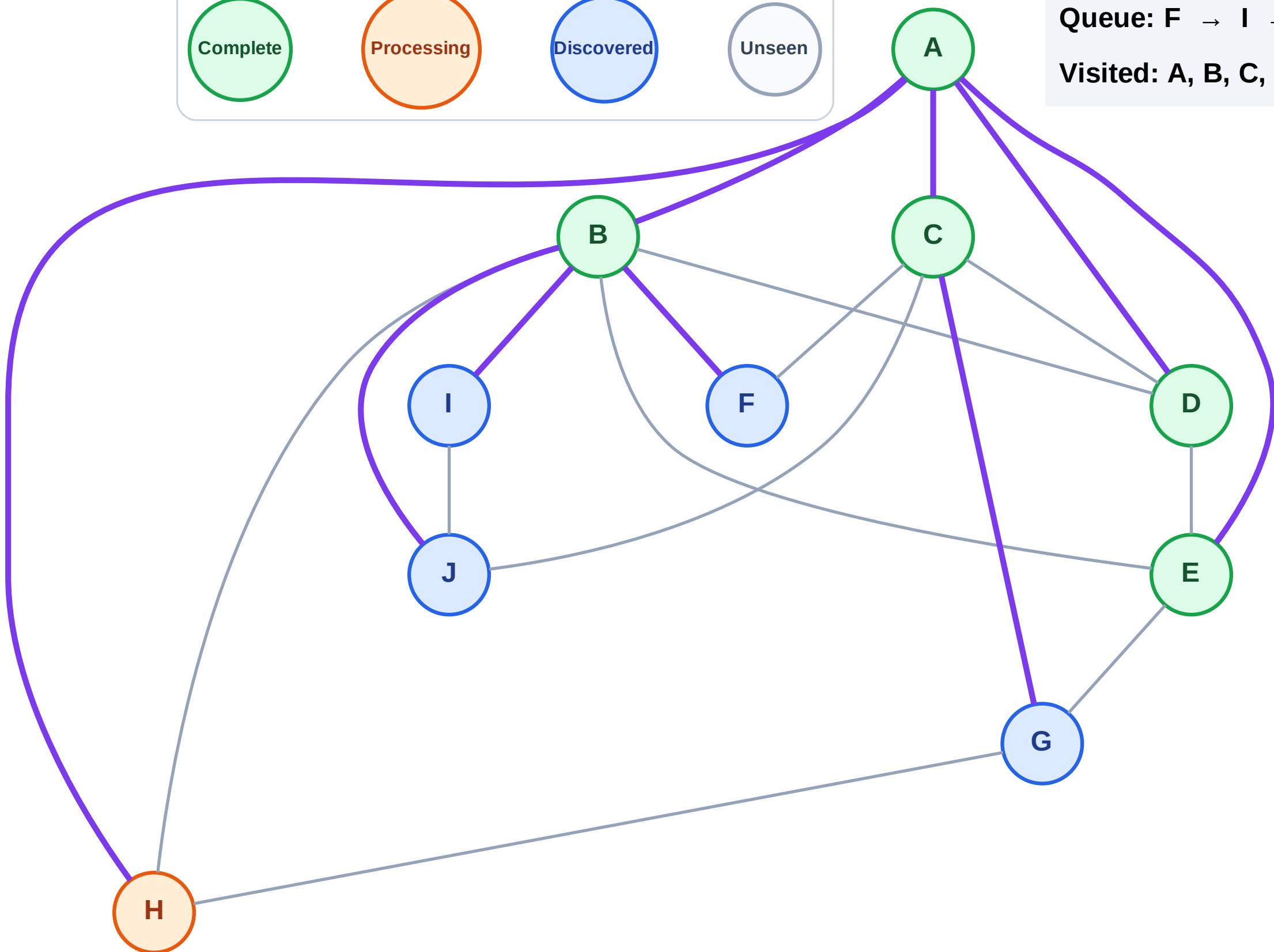
Step 12: Process node H

Legend



Queue: F → I → J → G

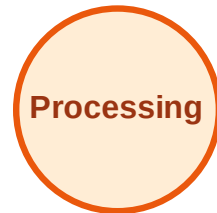
Visited: A, B, C, D, E



BFS Traversal

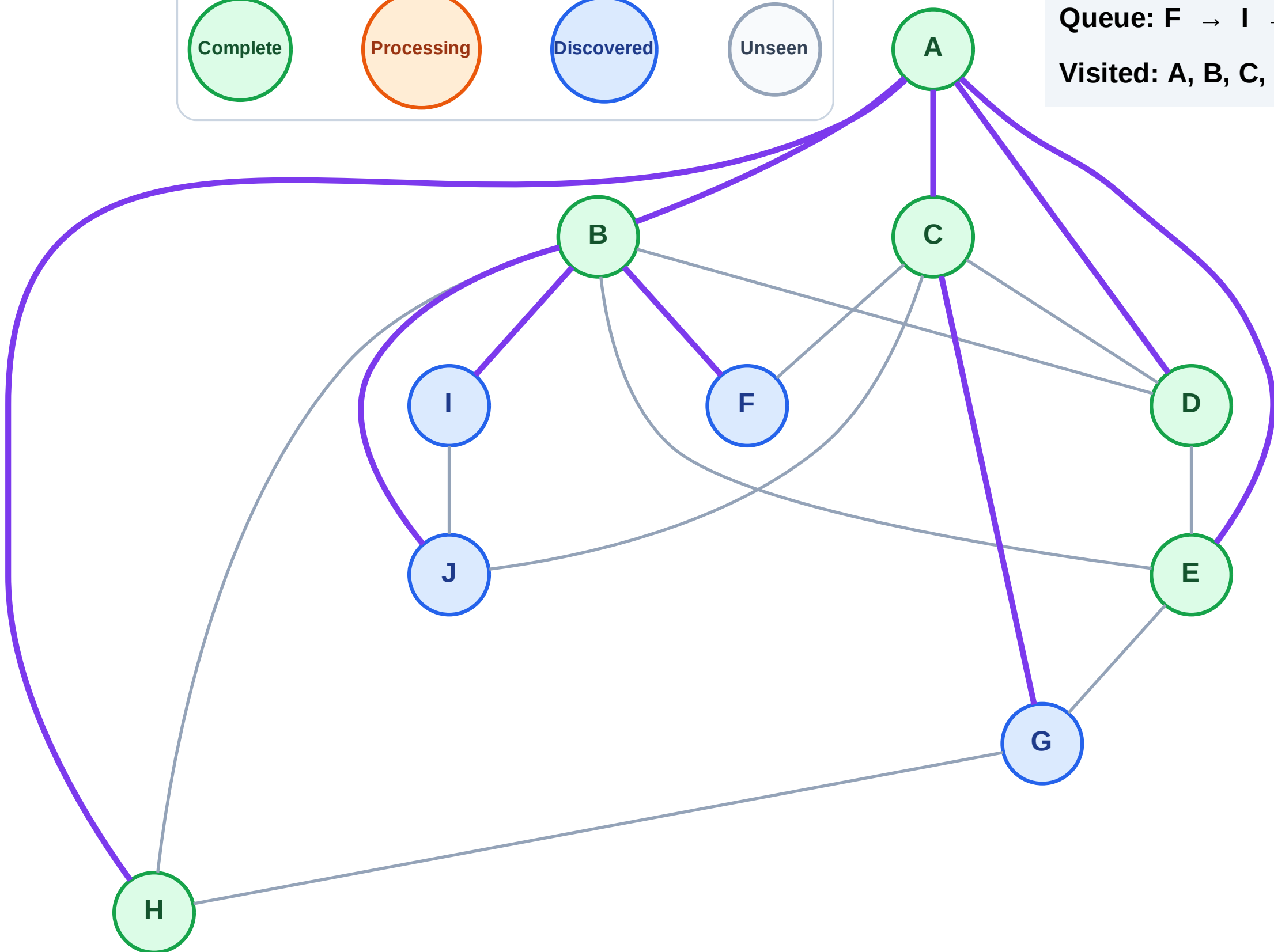
Step 13: No new neighbors from H

Legend



Queue: F → I → J → G

Visited: A, B, C, D, E, H



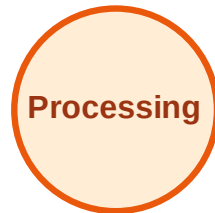
BFS Traversal

Step 14: Process node F

Legend



Complete



Processing



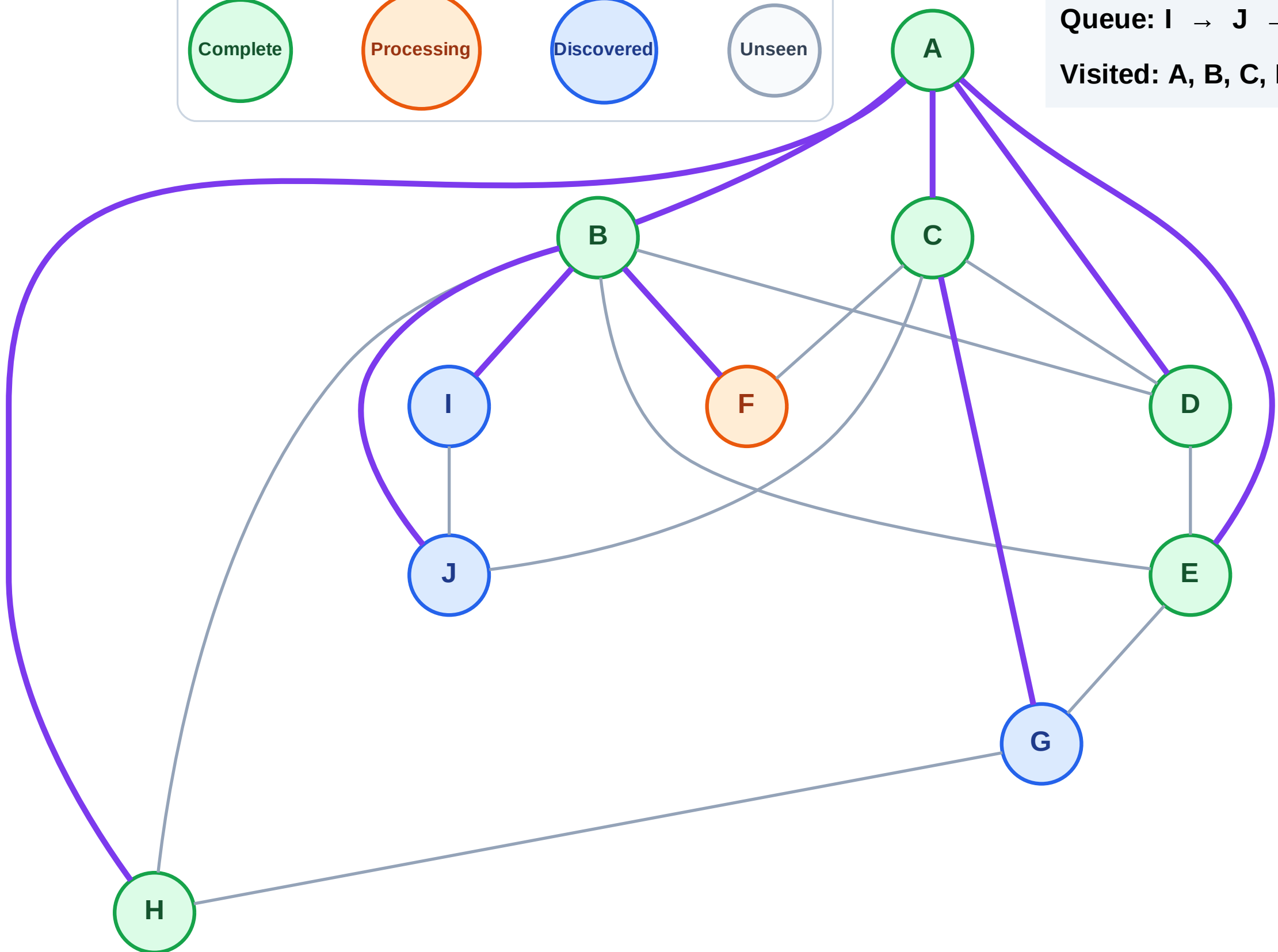
Discovered



Unseen

Queue: I → J → G

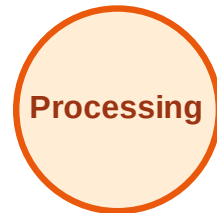
Visited: A, B, C, D, E, H



BFS Traversal

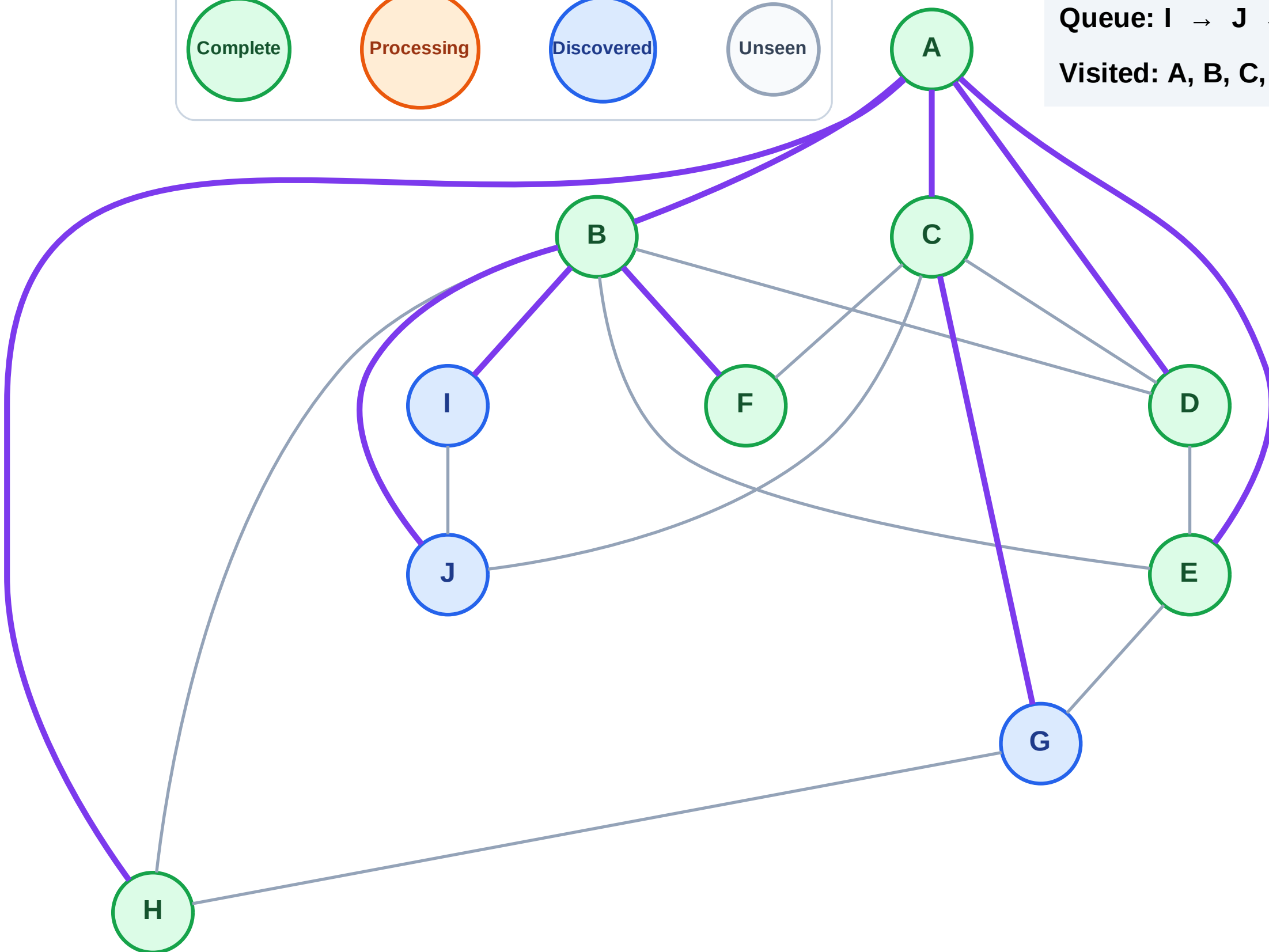
Step 15: No new neighbors from F

Legend



Queue: I → J → G

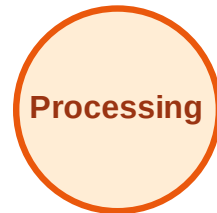
Visited: A, B, C, D, E, F, H



BFS Traversal

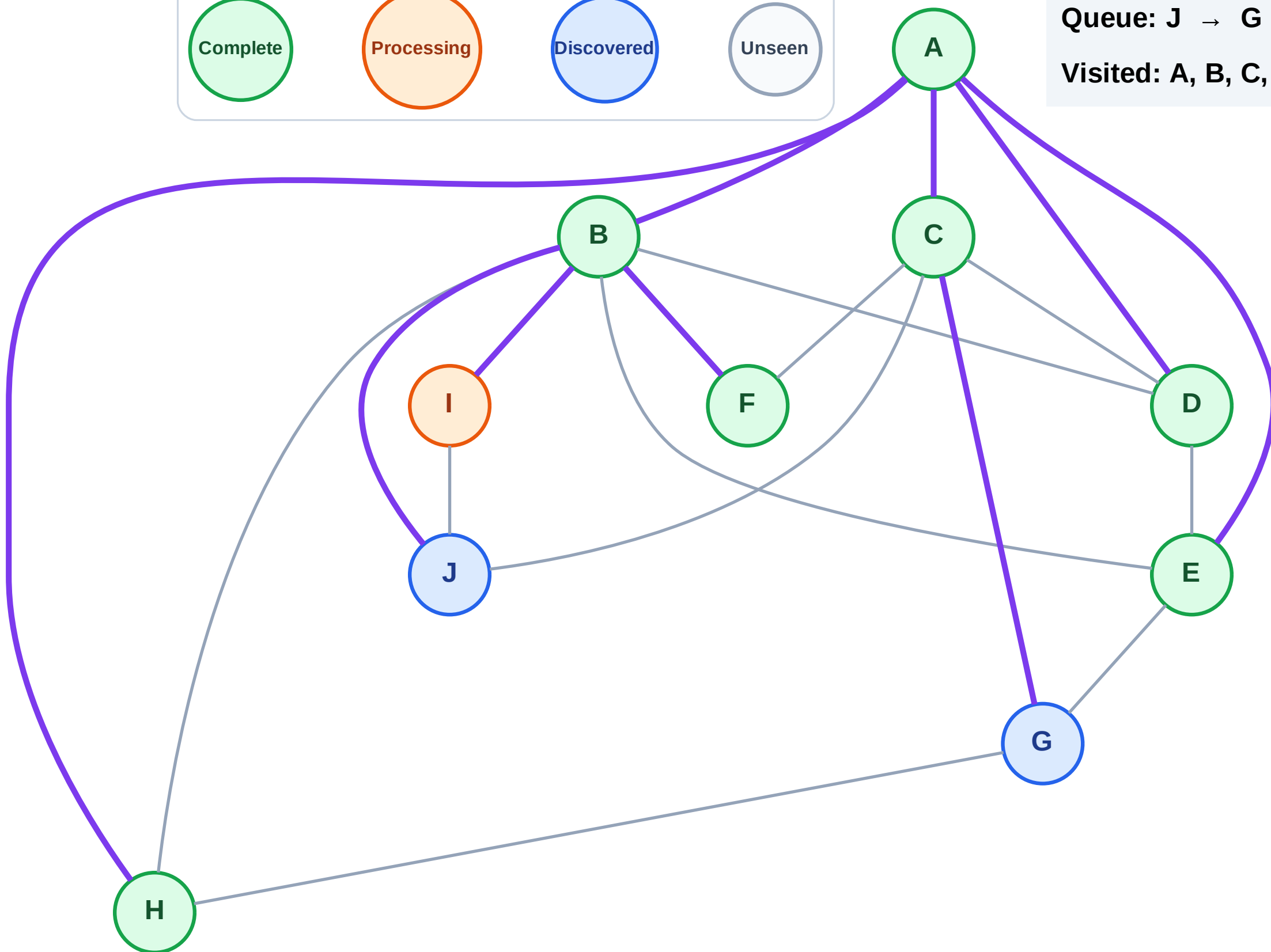
Step 16: Process node I

Legend



Queue: J → G

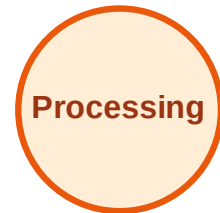
Visited: A, B, C, D, E, F, H



BFS Traversal

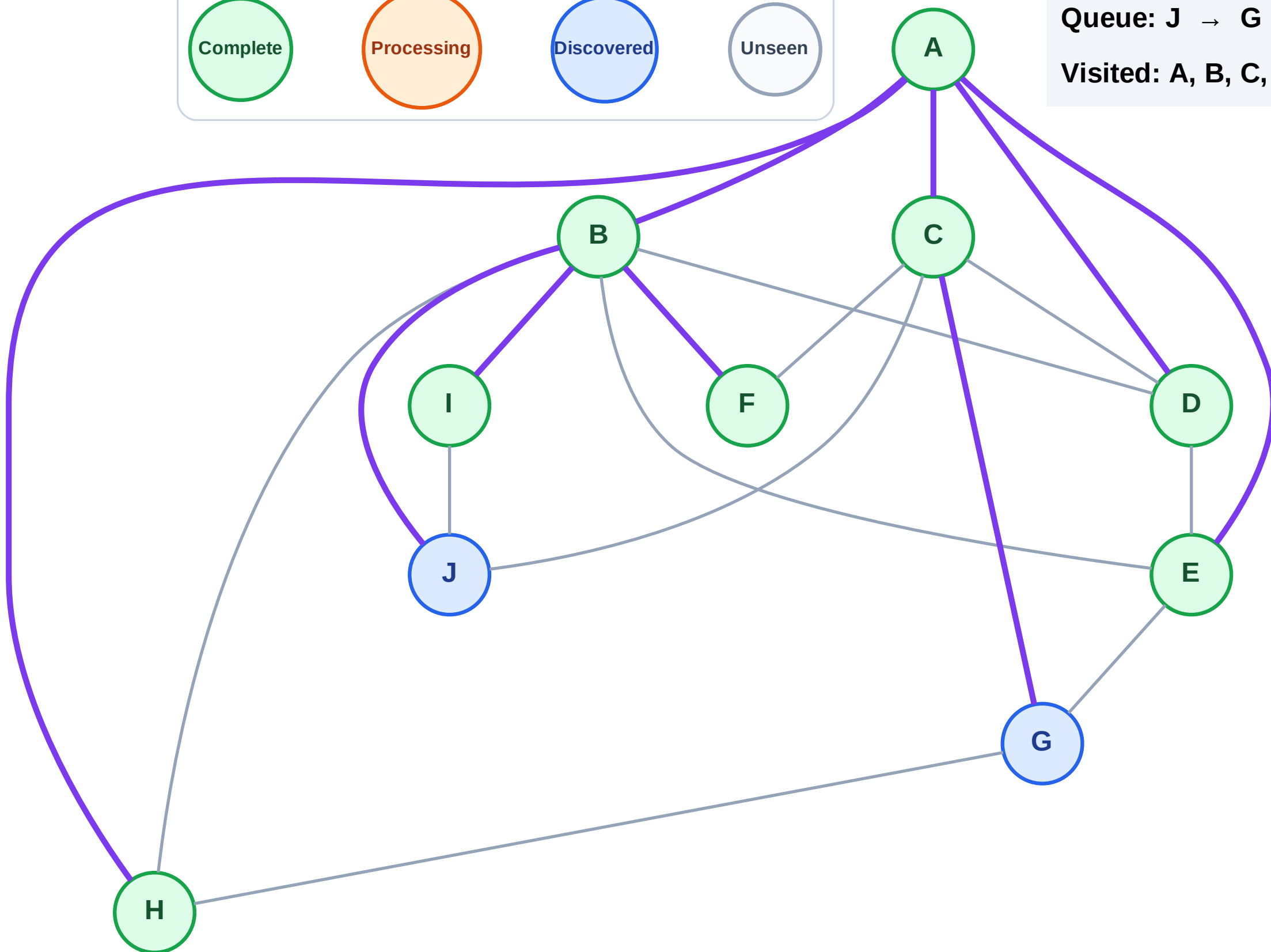
Step 17: No new neighbors from I

Legend



Queue: J → G

Visited: A, B, C, D, E, F, H, I



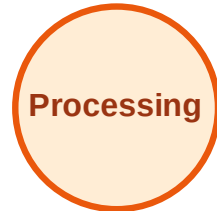
BFS Traversal

Step 18: Process node J

Legend



Complete



Processing



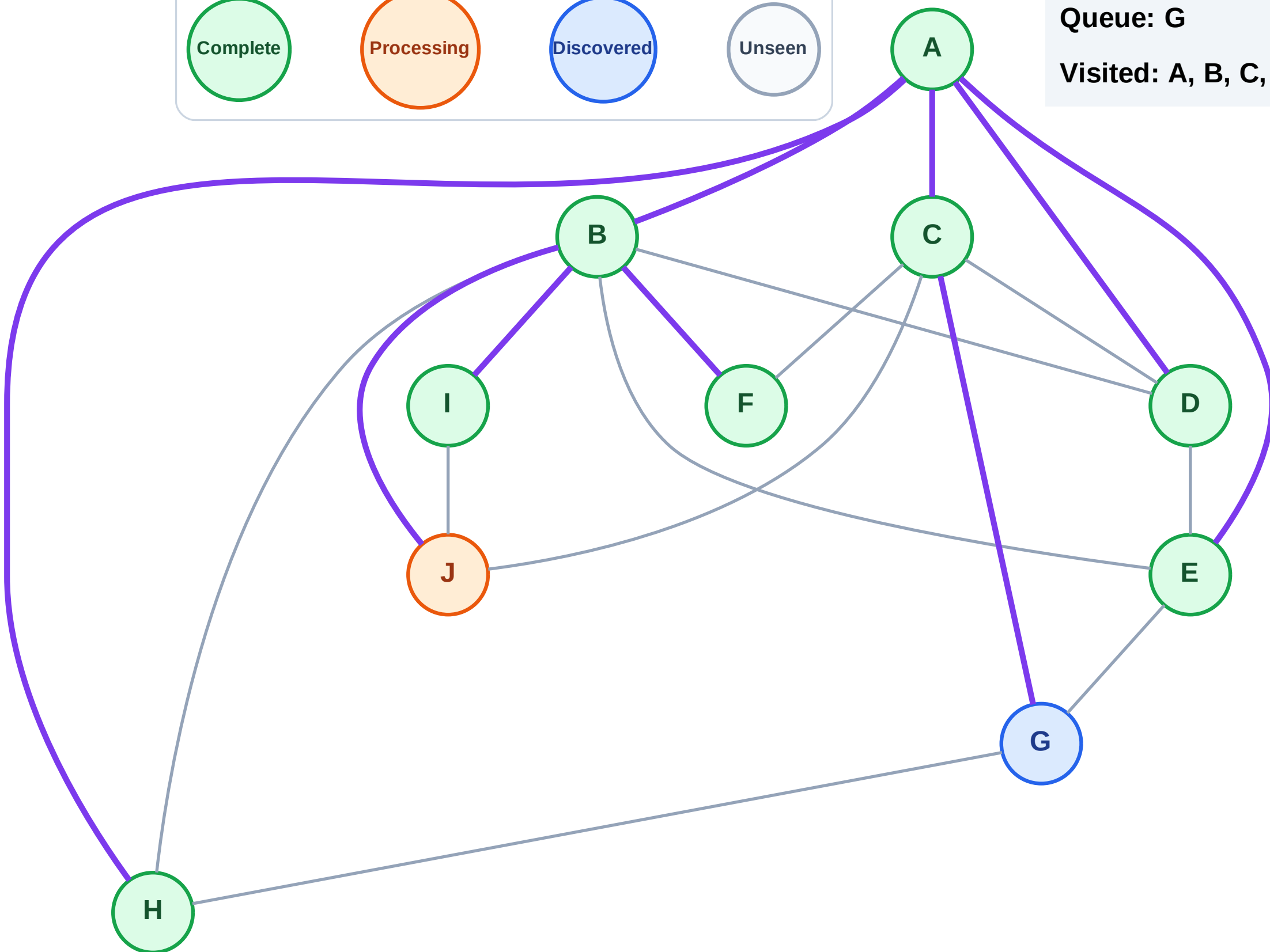
Discovered



Unseen

Queue: G

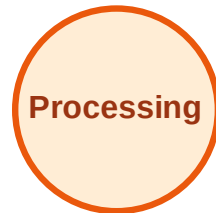
Visited: A, B, C, D, E, F, H, I



BFS Traversal

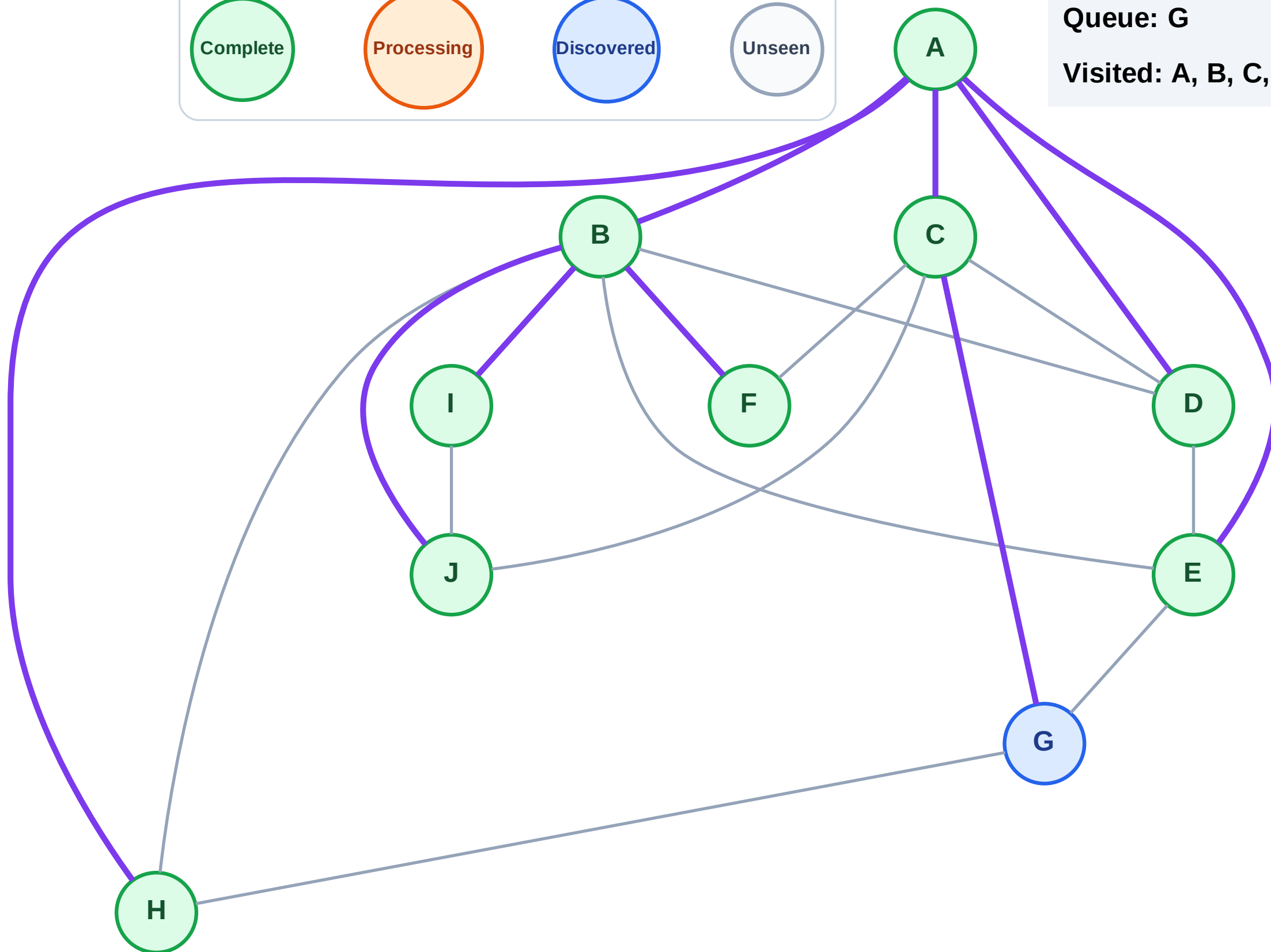
Step 19: No new neighbors from J

Legend



Queue: G

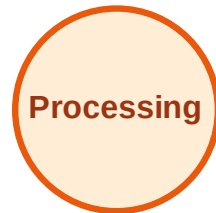
Visited: A, B, C, D, E, F, H, I, J



BFS Traversal

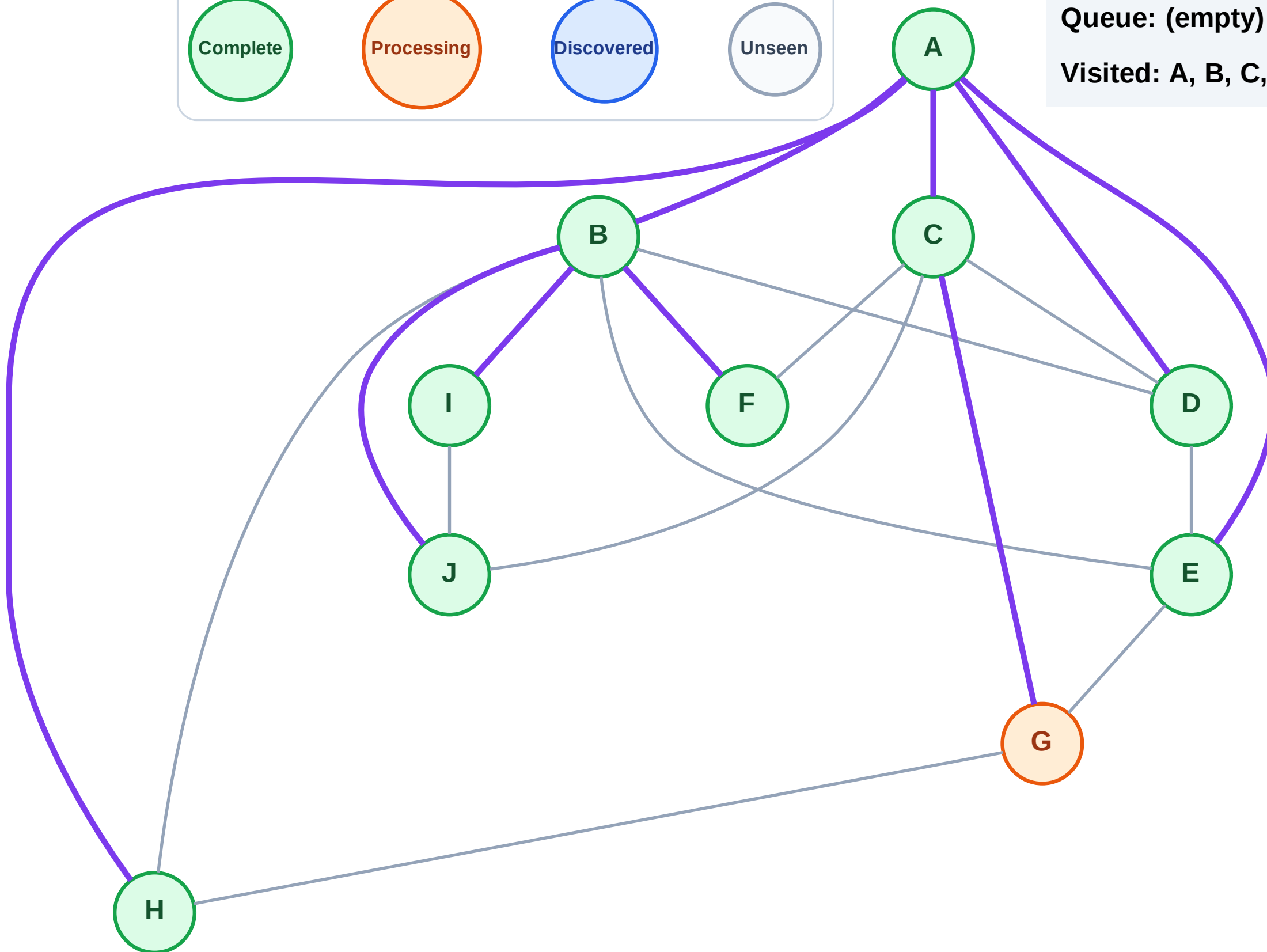
Step 20: Process node G

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, H, I, J



Step 21: No new neighbors from G

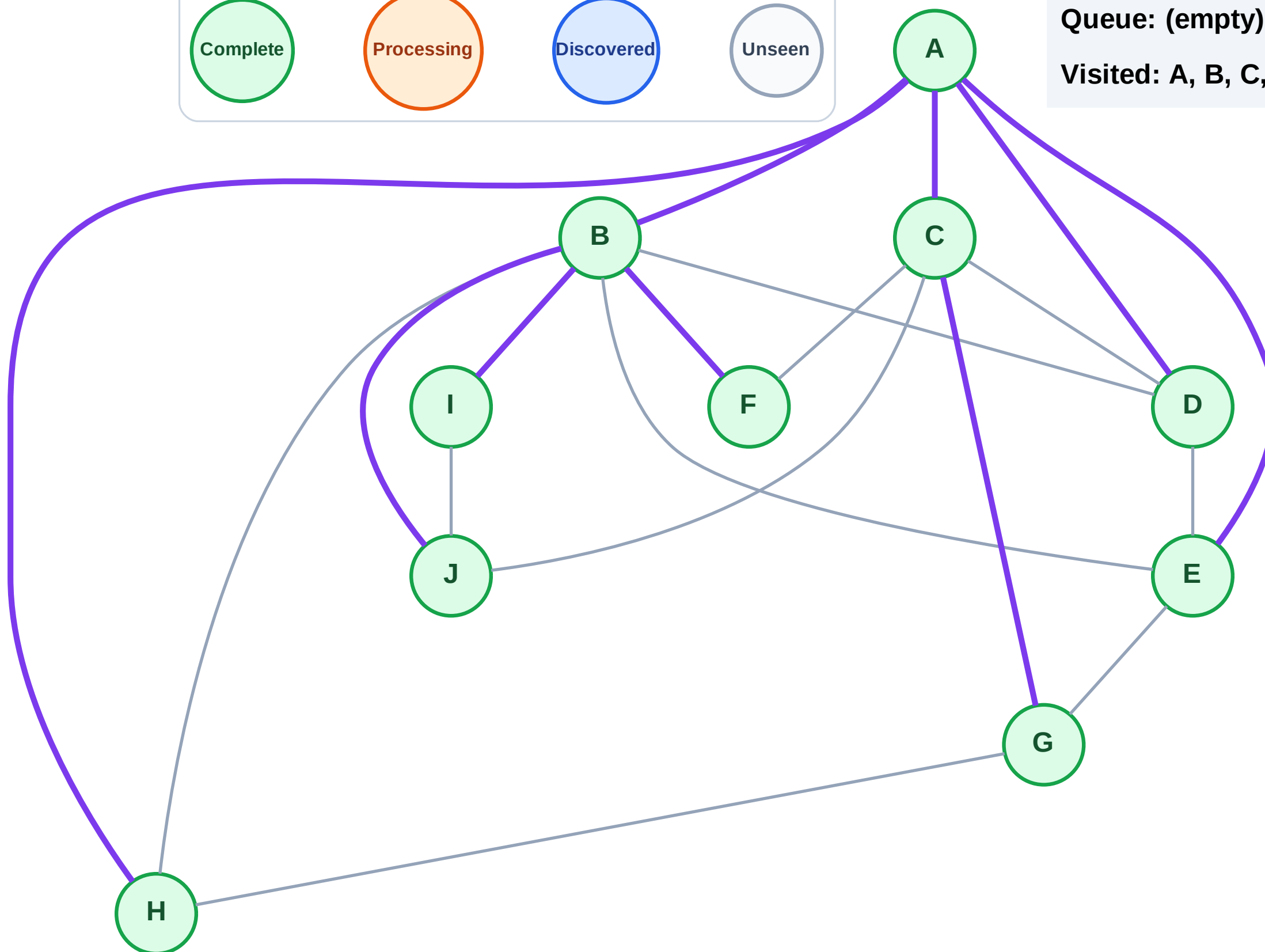
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle with the label 'Complete', an orange circle with the label 'Processing', a blue circle with the label 'Discovered', and a grey circle with the label 'Unseen'.

- Complete
- Processing
- Discovered
- Unseen

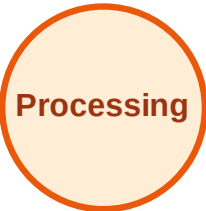
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

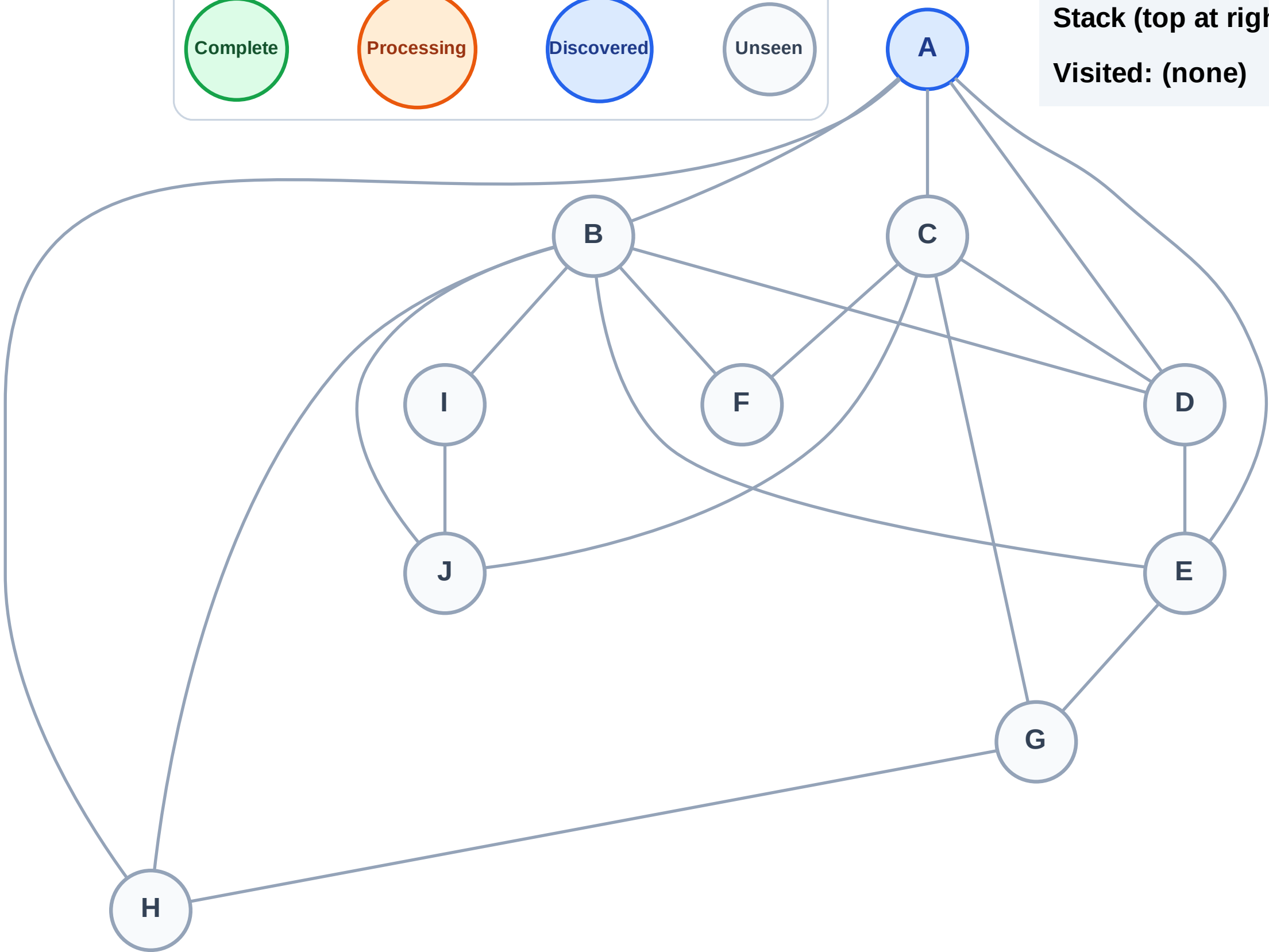
Step 1: Start at A

Legend



Stack (top at right): A

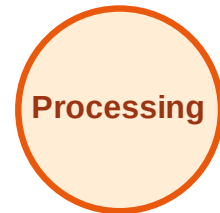
Visited: (none)



DFS Traversal

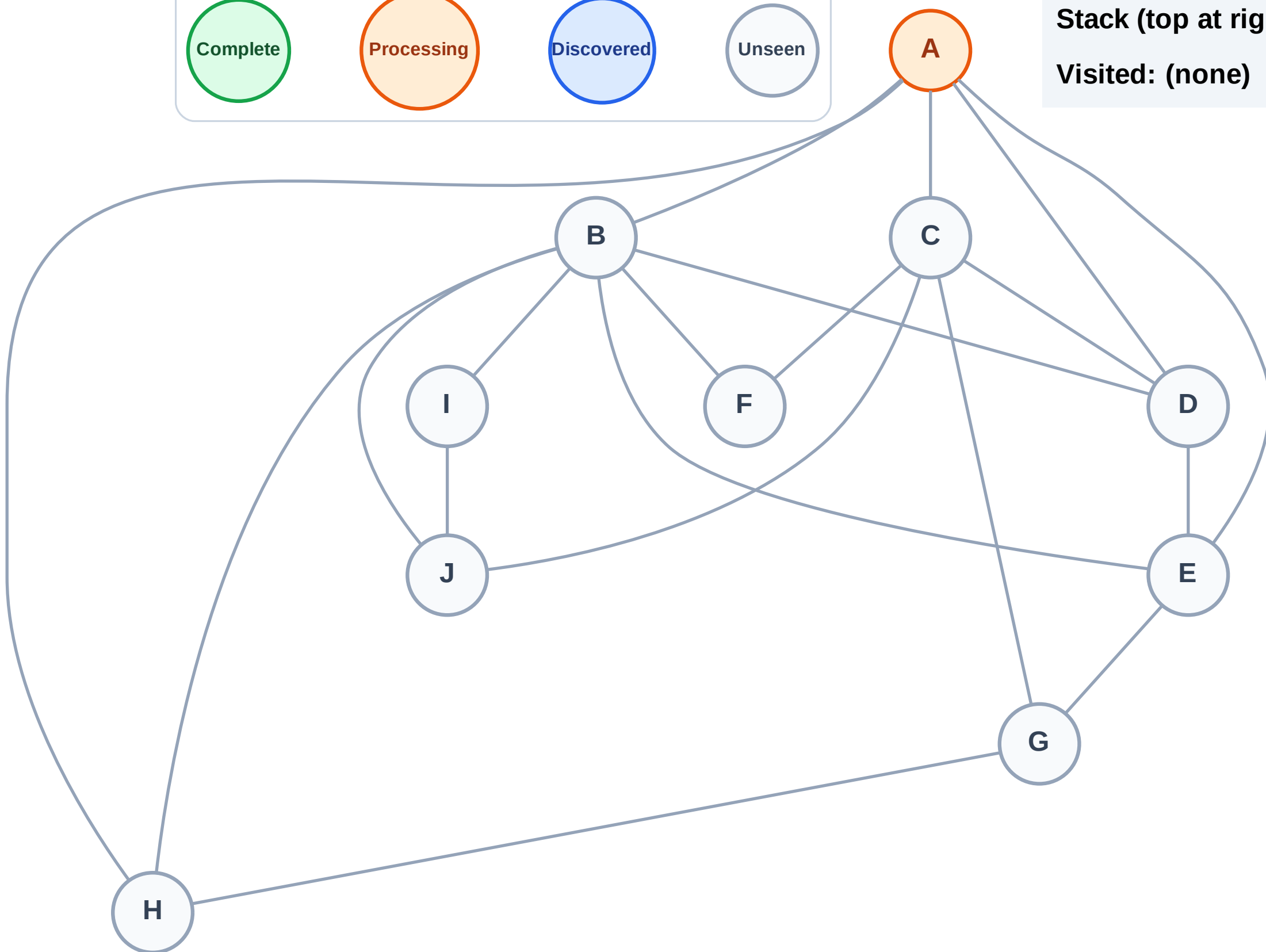
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

Step 3: Discover H, E, D, C, B from A

Legend

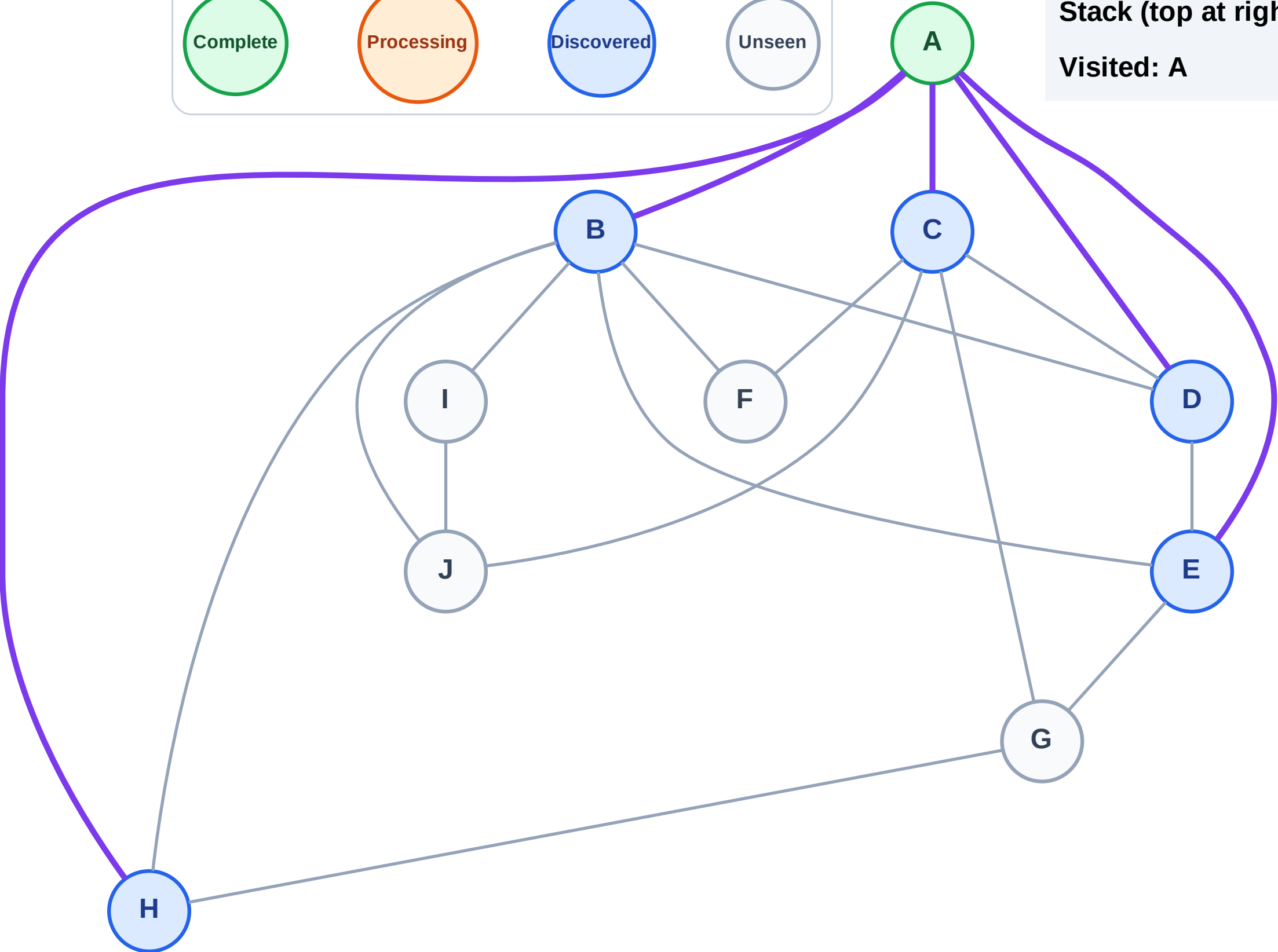
Complete

Processing

Discovered

Unseen

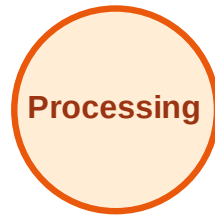
Stack (top at right): H | E | D | C | B
Visited: A



DFS Traversal

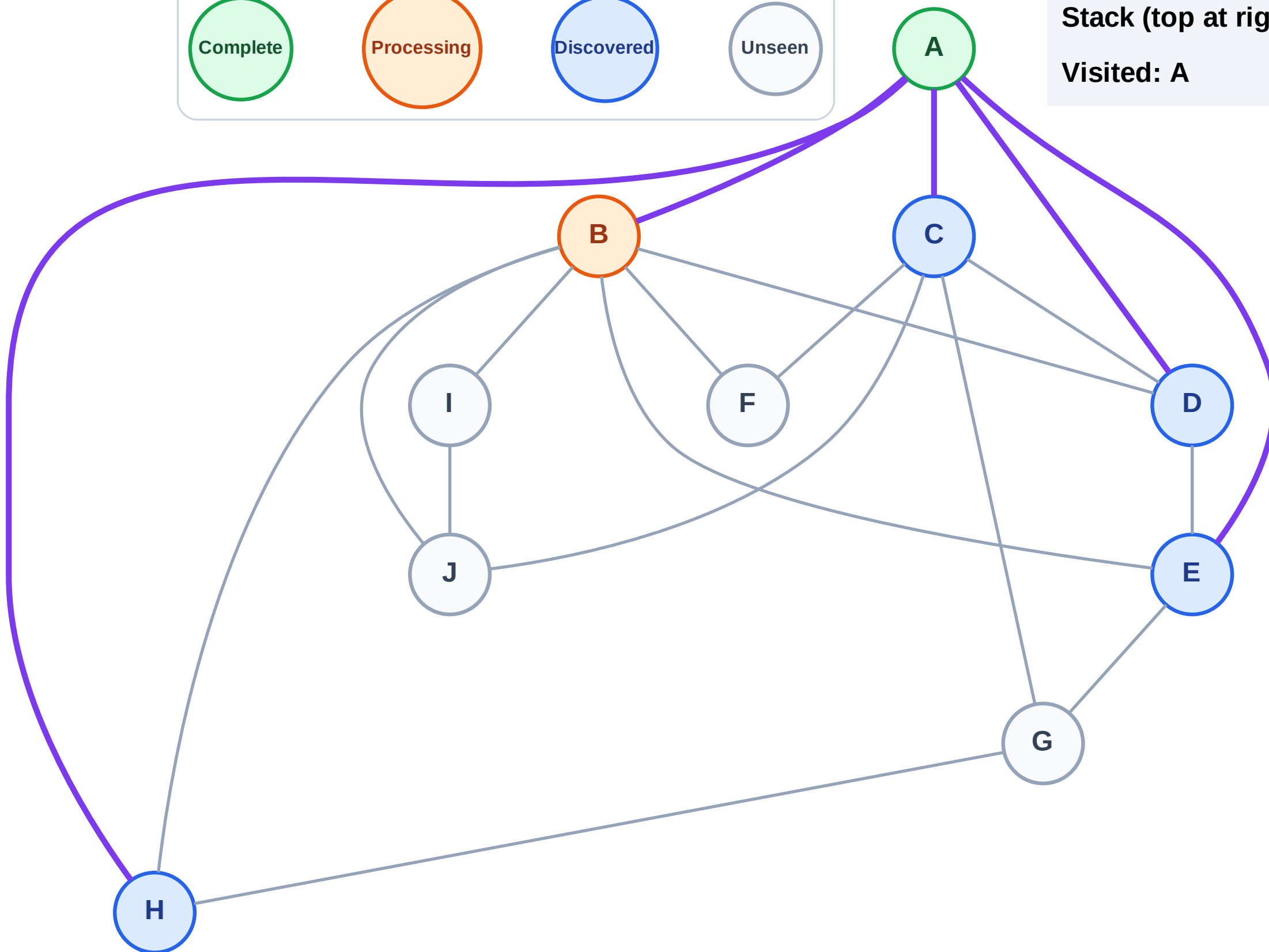
Step 4: Process node B

Legend



Stack (top at right): H | E | D | C

Visited: A



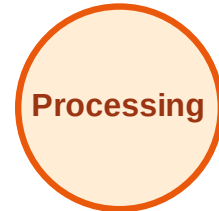
DFS Traversal

Step 5: Discover J, I, F from B

Legend



Complete



Processing



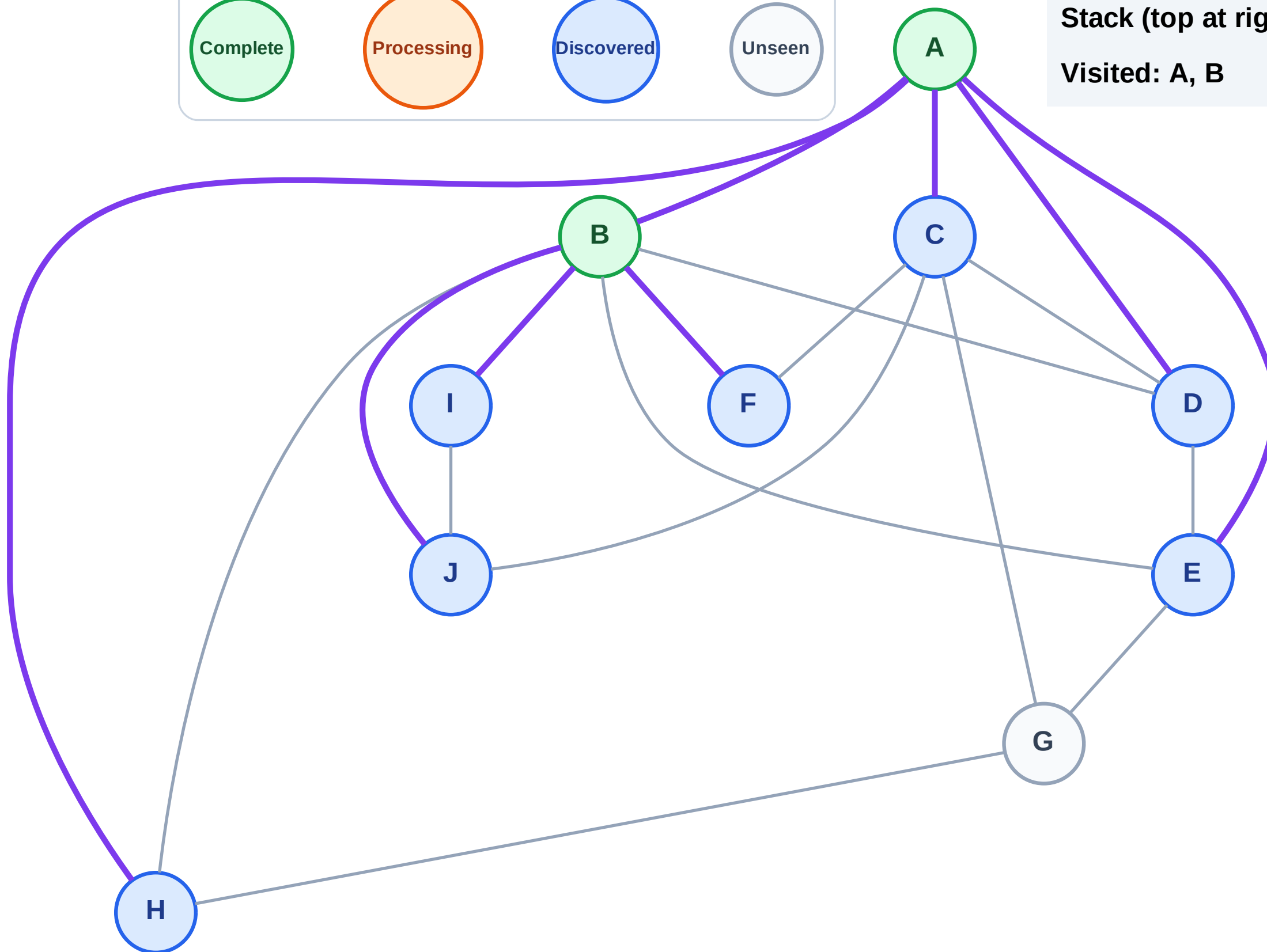
Discovered



Unseen

Stack (top at right): H | E | D | C | J | I | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

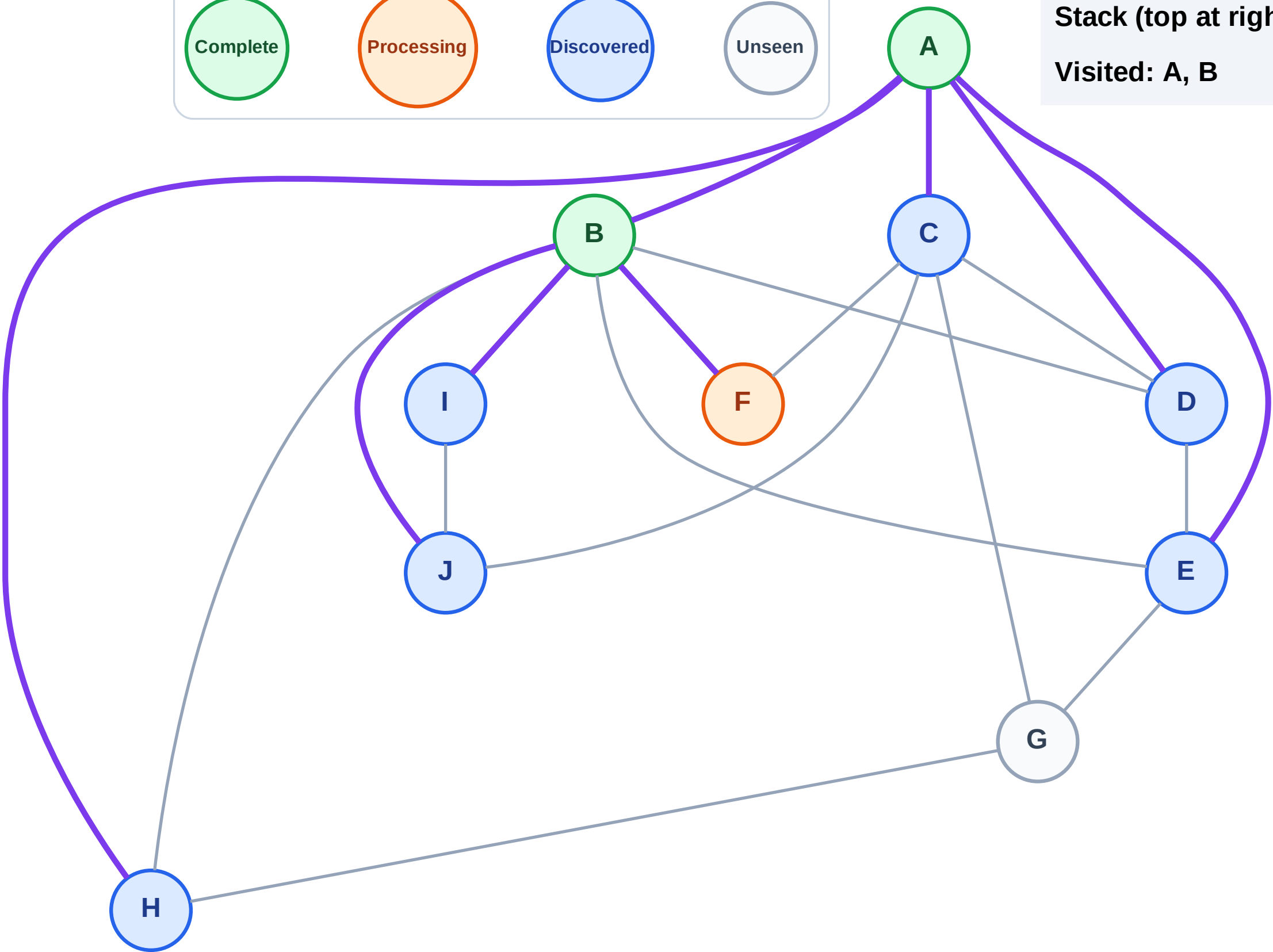
Processing

Discovered

Unseen

Stack (top at right): H | E | D | C | J | I

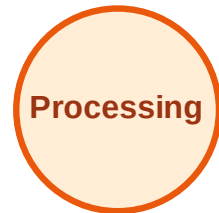
Visited: A, B



DFS Traversal

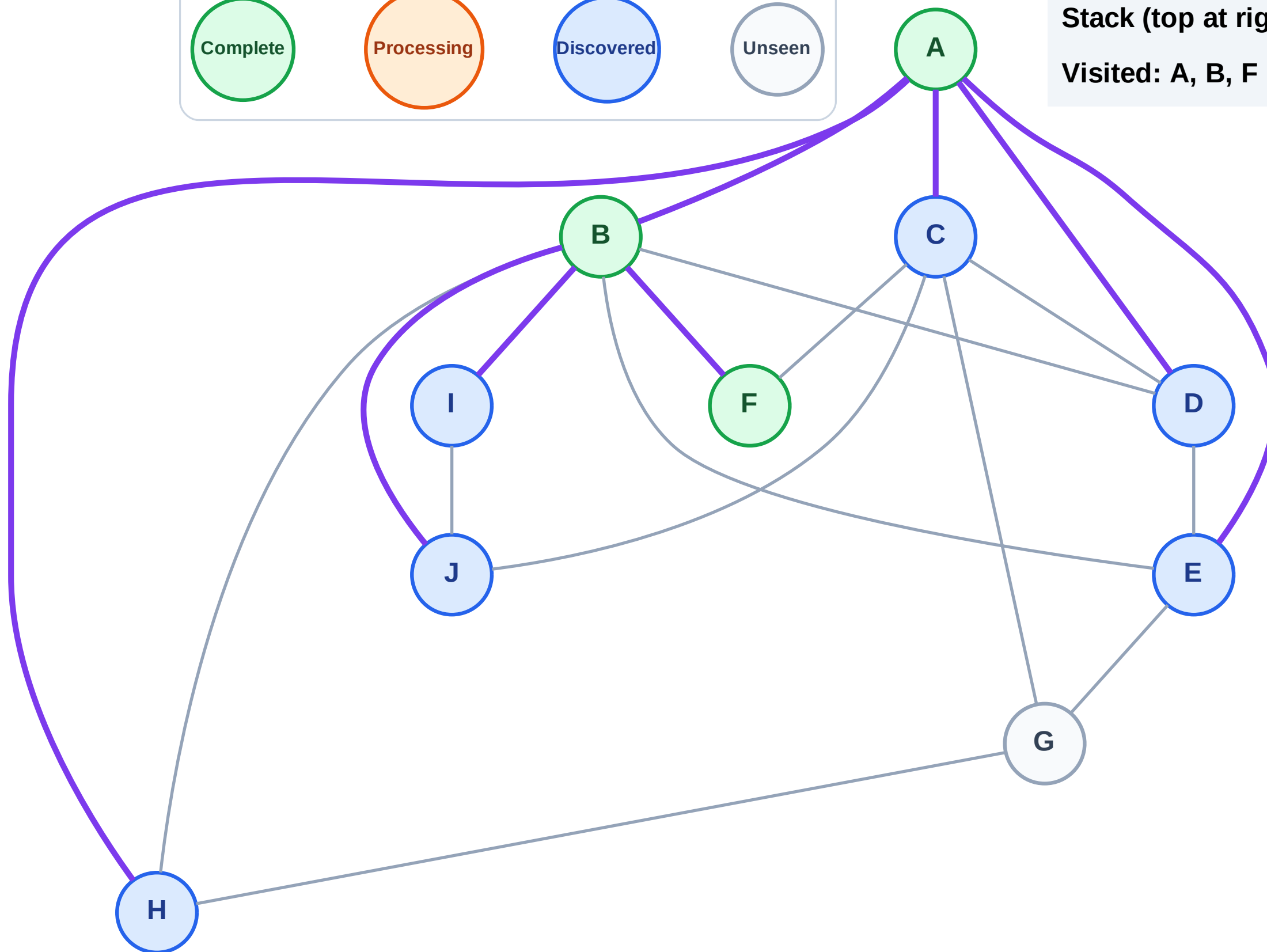
Step 7: No new neighbors from F

Legend



Stack (top at right): H | E | D | C | J | I

Visited: A, B, F



DFS Traversal

Step 8: Process node I

Legend

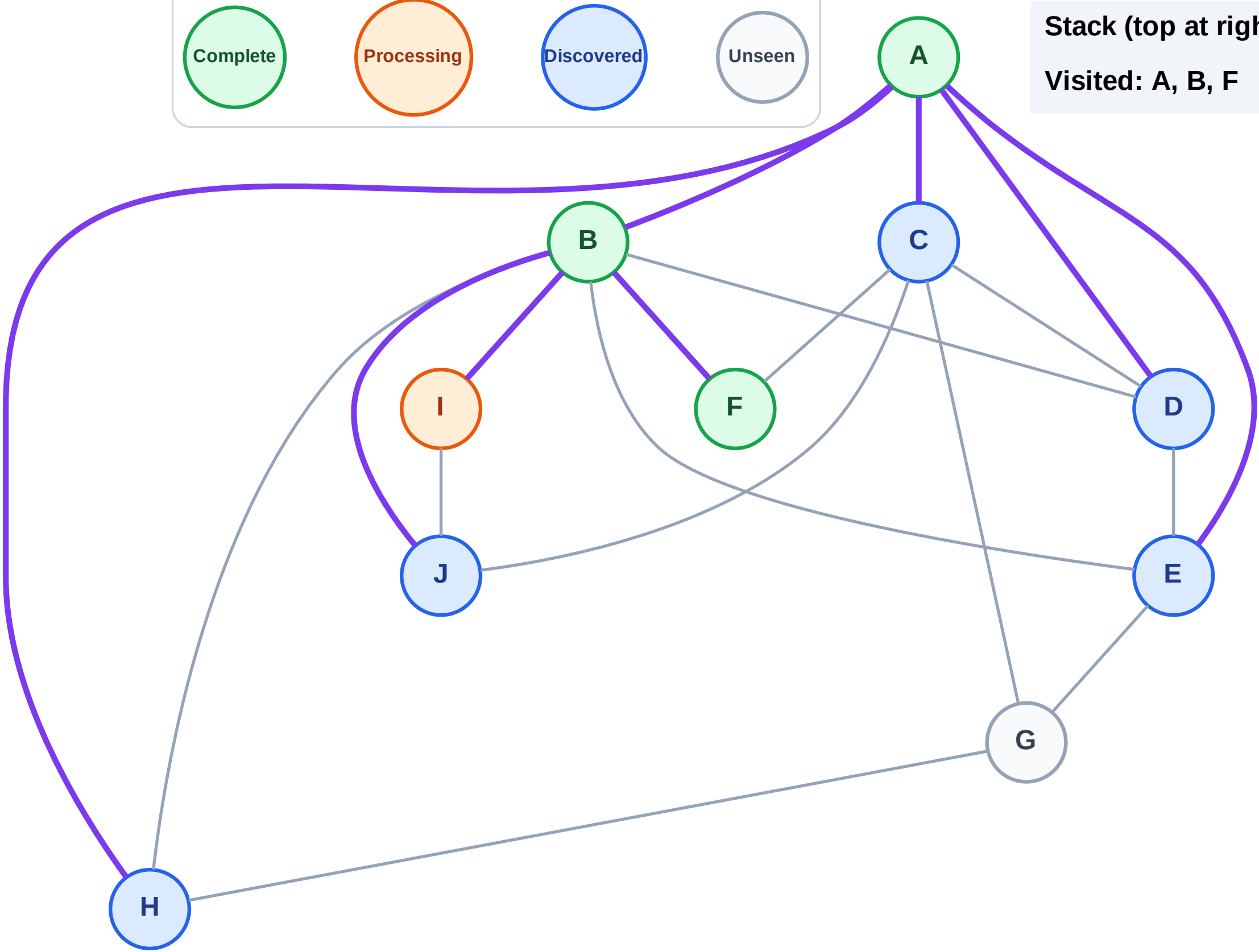
Complete

Processing

Discovered

Unseen

Stack (top at right): H | E | D | C | J
Visited: A, B, F



DFS Traversal

Step 9: No new neighbors from I

Legend

Complete

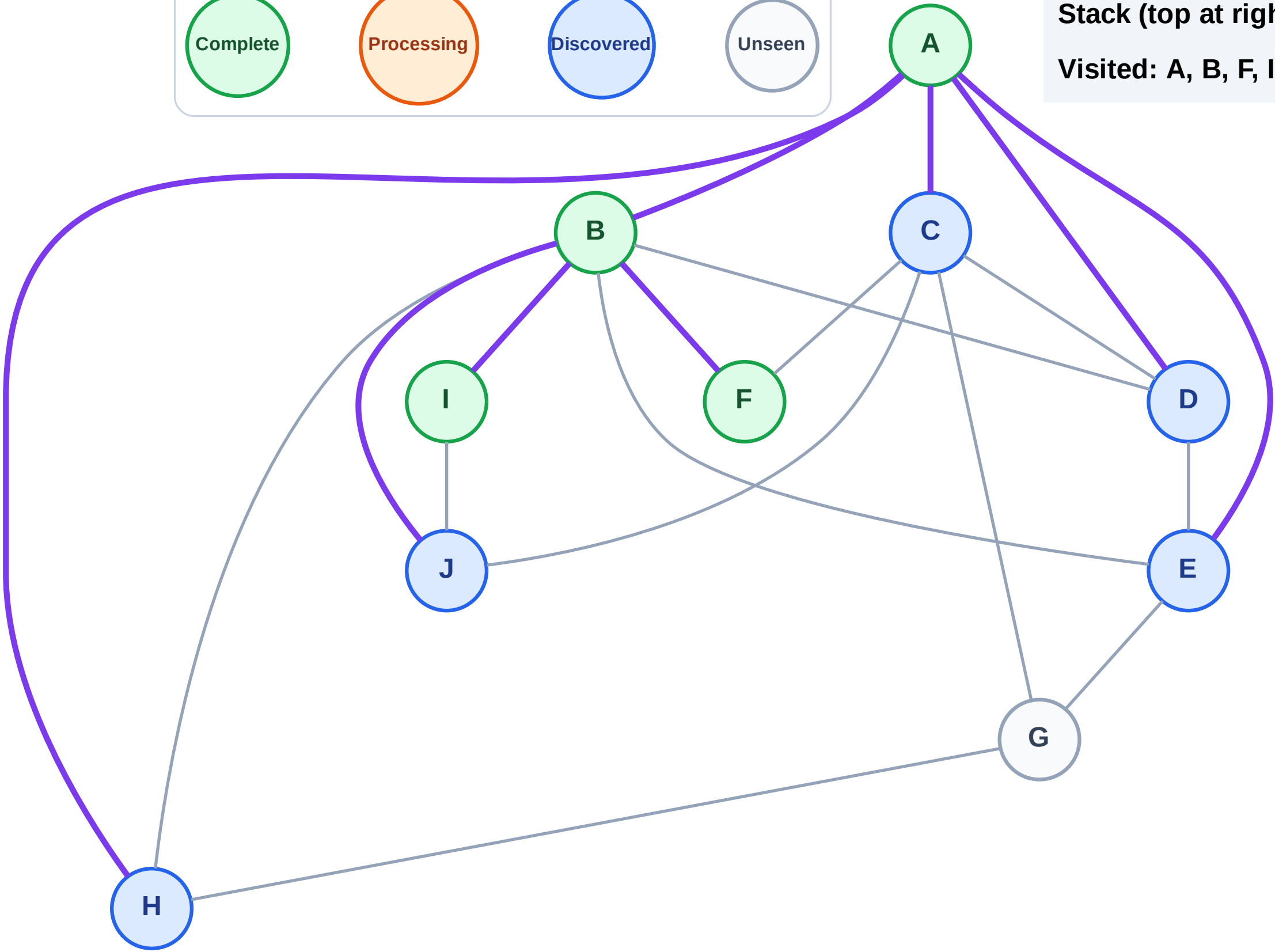
Processing

Discovered

Unseen

Stack (top at right): H | E | D | C | J

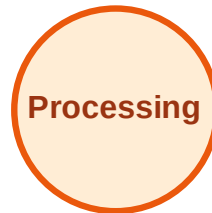
Visited: A, B, F, I



DFS Traversal

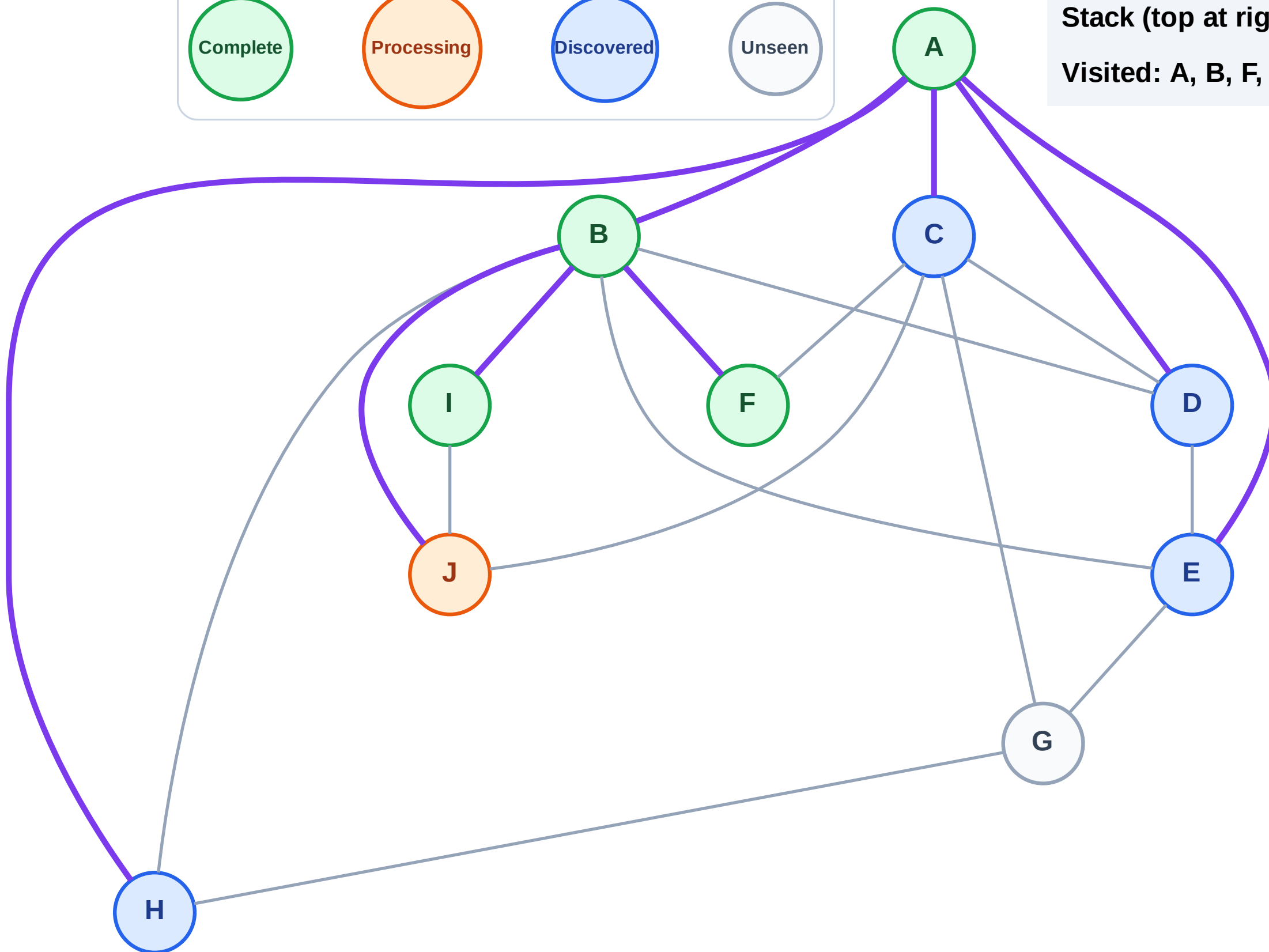
Step 10: Process node J

Legend



Stack (top at right): H | E | D | C

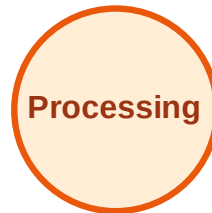
Visited: A, B, F, I



DFS Traversal

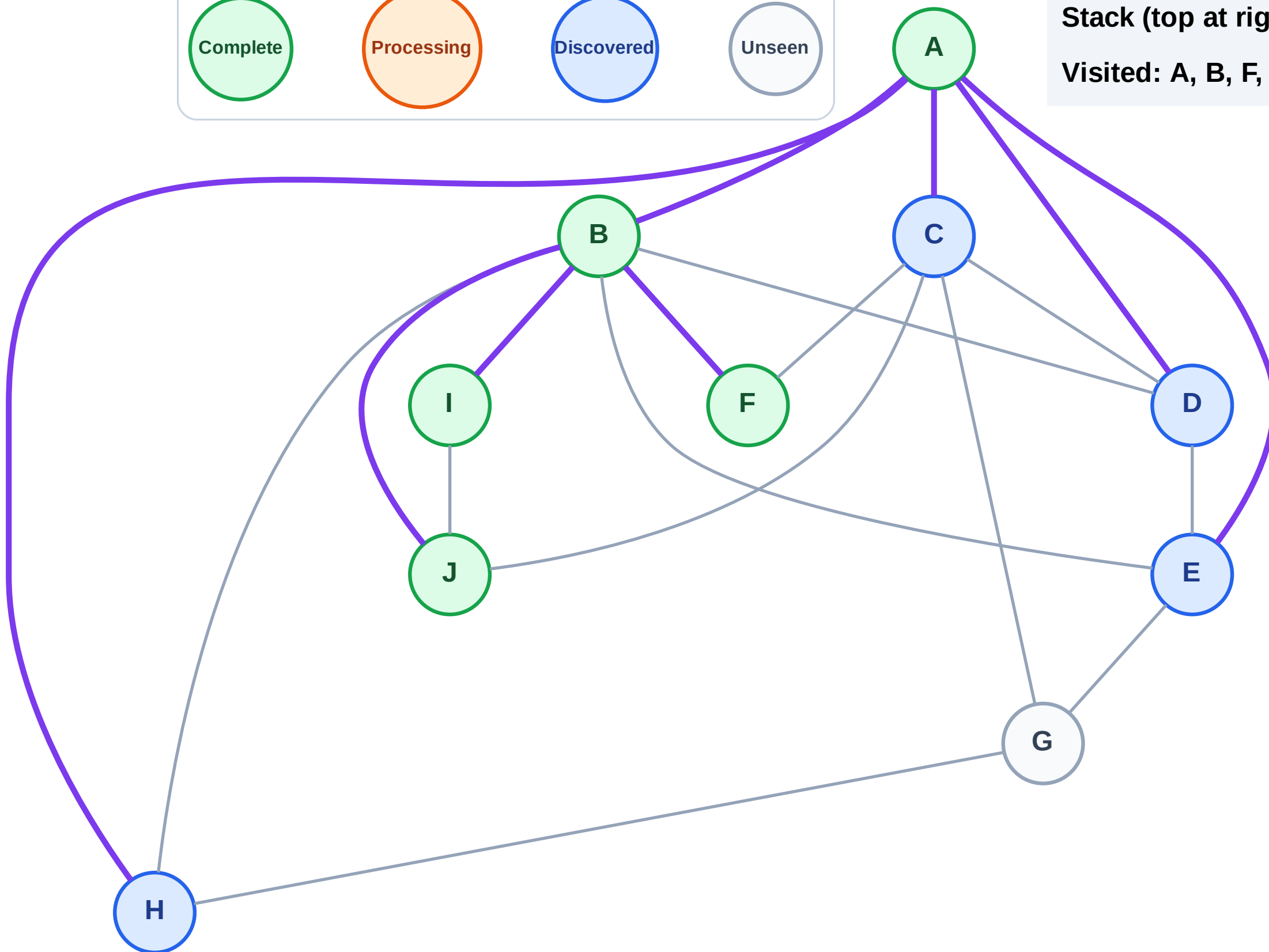
Step 11: No new neighbors from J

Legend



Stack (top at right): H | E | D | C

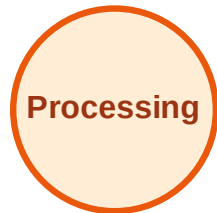
Visited: A, B, F, I, J



DFS Traversal

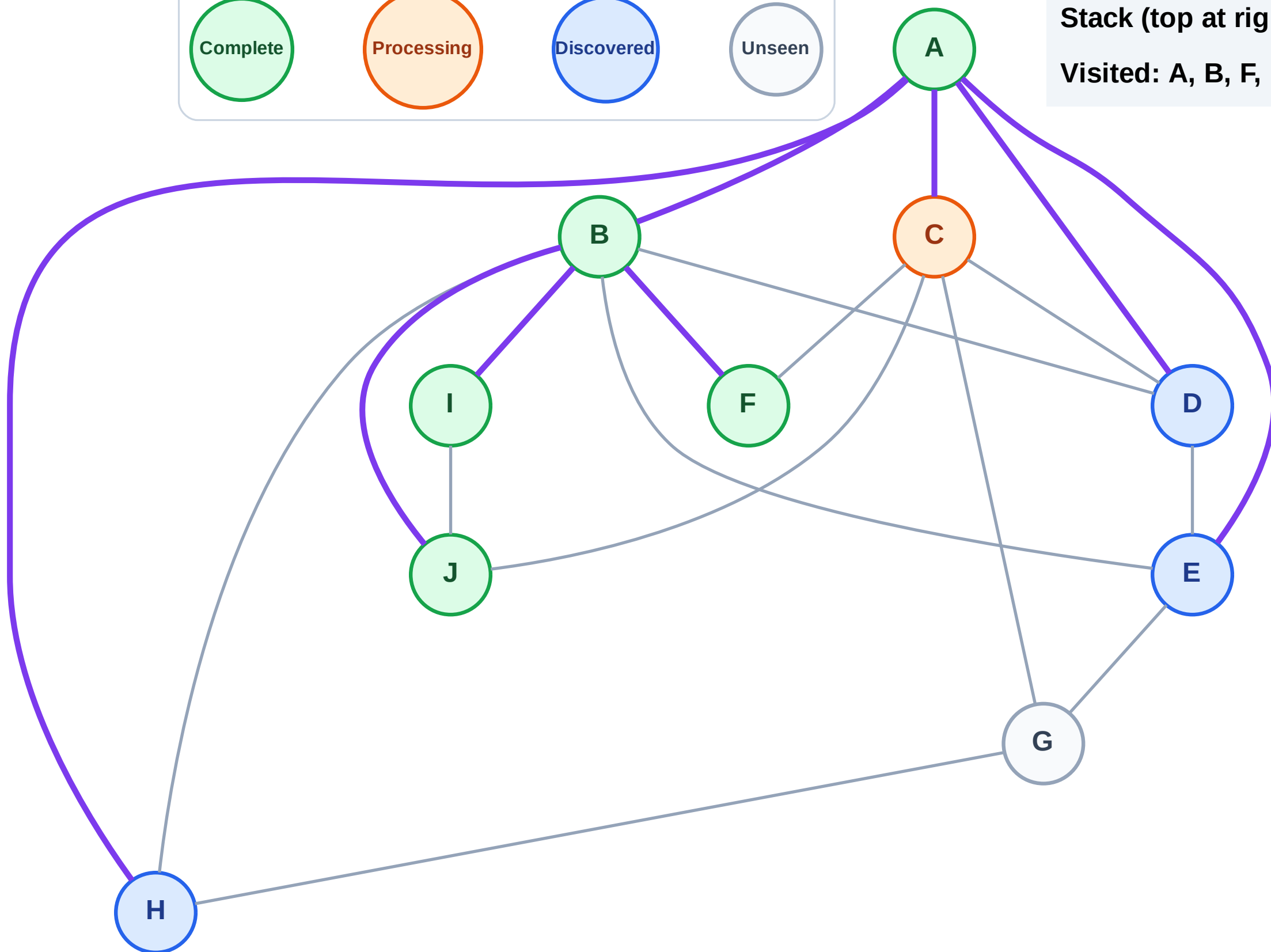
Step 12: Process node C

Legend



Stack (top at right): H | E | D

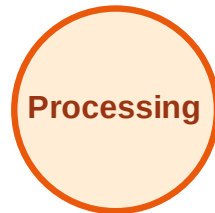
Visited: A, B, F, I, J



DFS Traversal

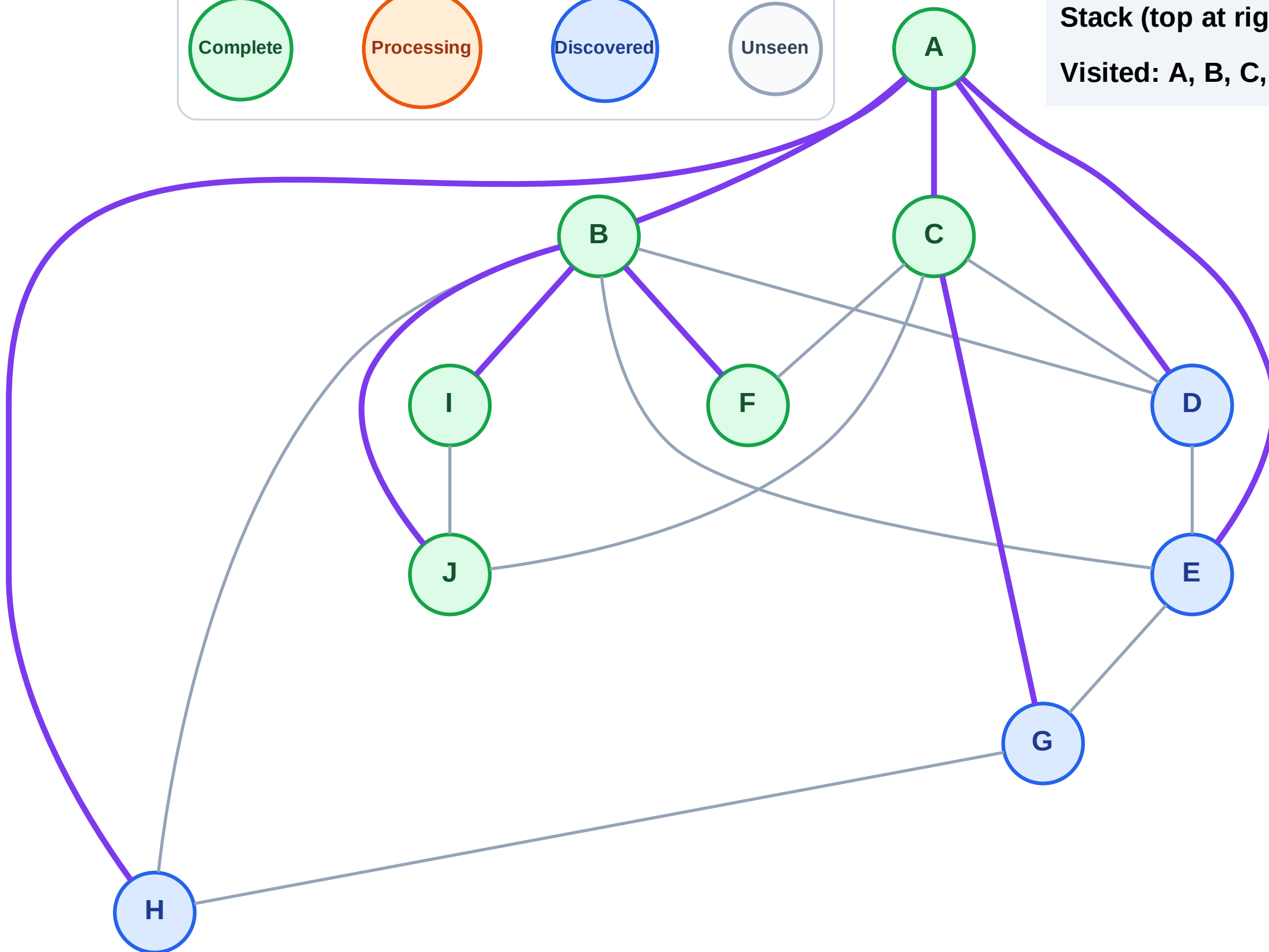
Step 13: Discover G from C

Legend



Stack (top at right): H | E | D | G

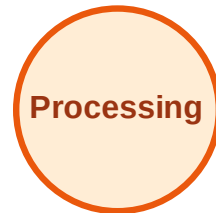
Visited: A, B, C, F, I, J



DFS Traversal

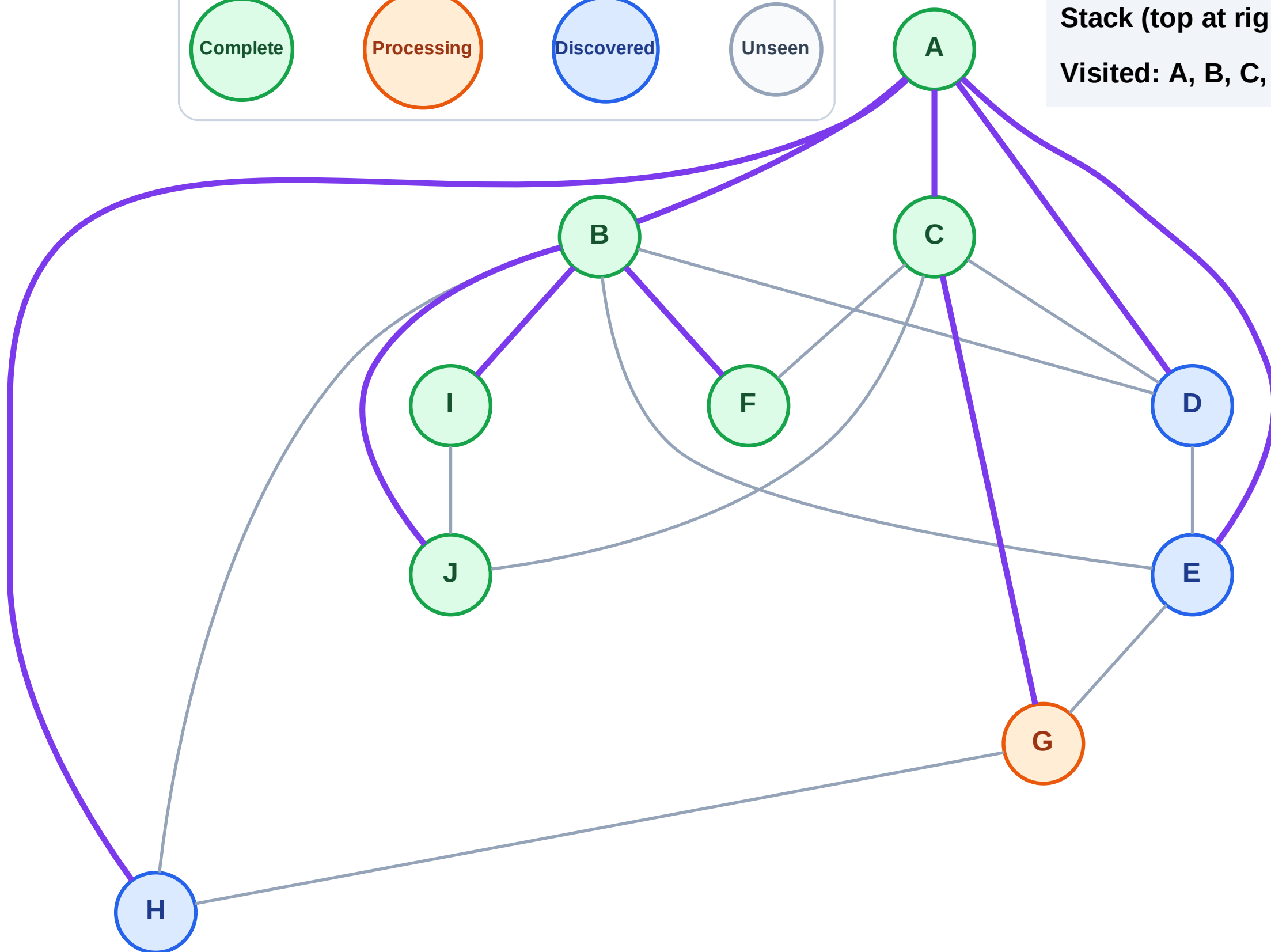
Step 14: Process node G

Legend



Stack (top at right): H | E | D

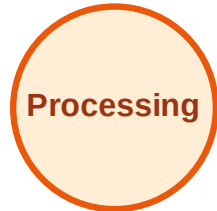
Visited: A, B, C, F, I, J



DFS Traversal

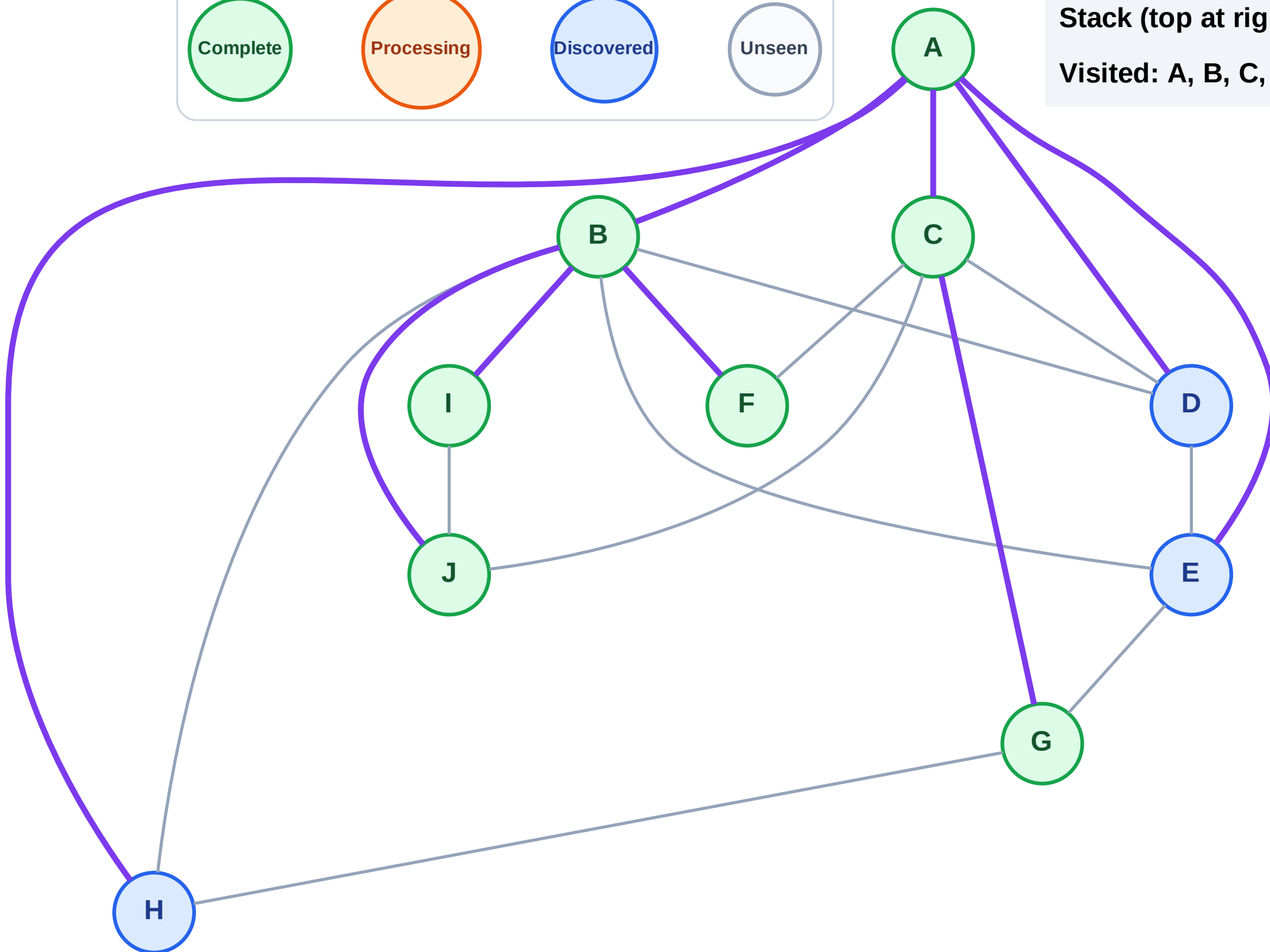
Step 15: No new neighbors from G

Legend



Stack (top at right): H | E | D

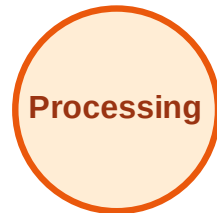
Visited: A, B, C, F, G, I, J



DFS Traversal

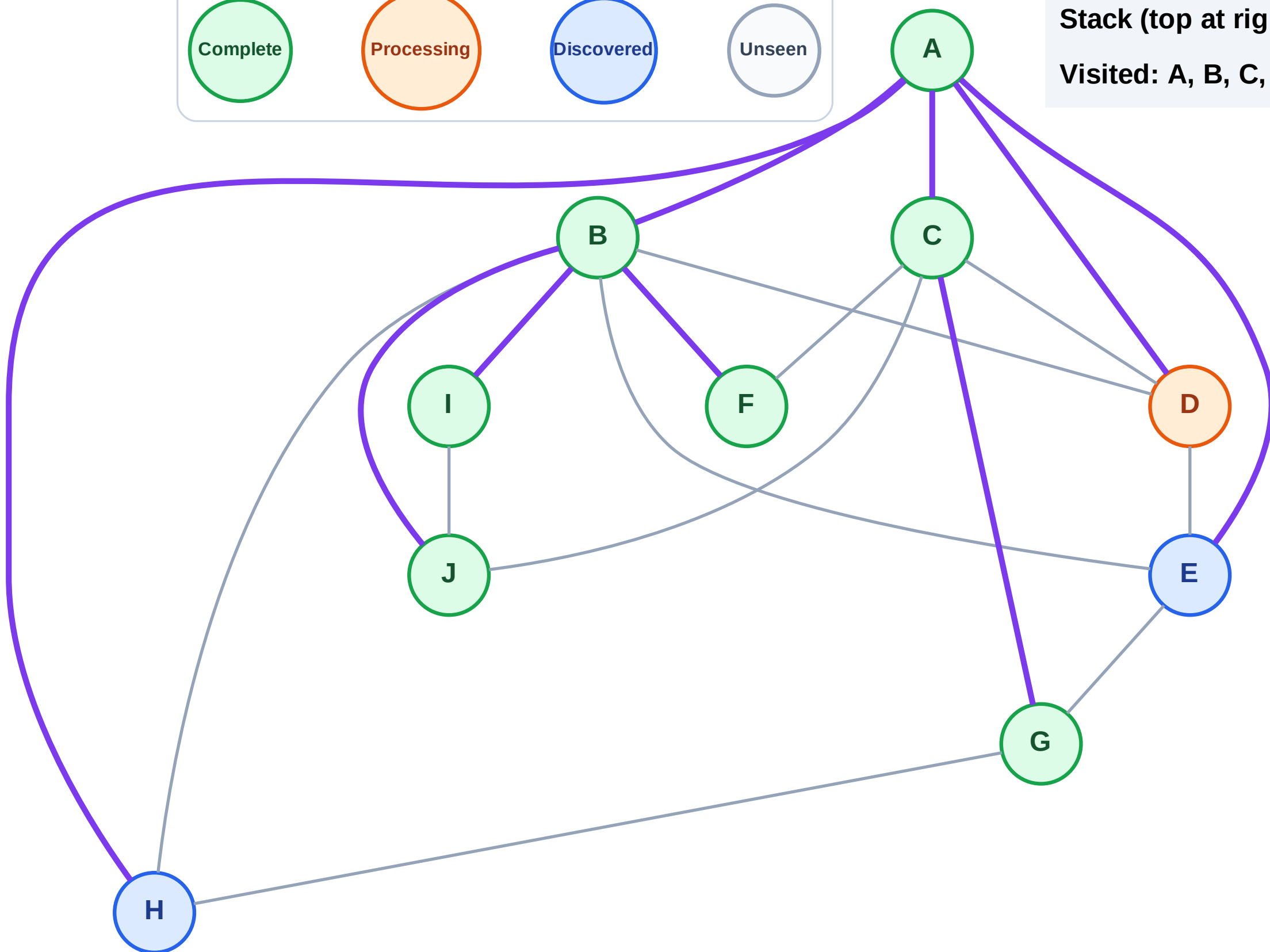
Step 16: Process node D

Legend



Stack (top at right): H | E

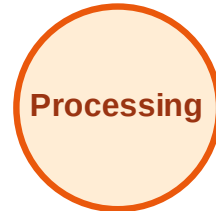
Visited: A, B, C, F, G, I, J



DFS Traversal

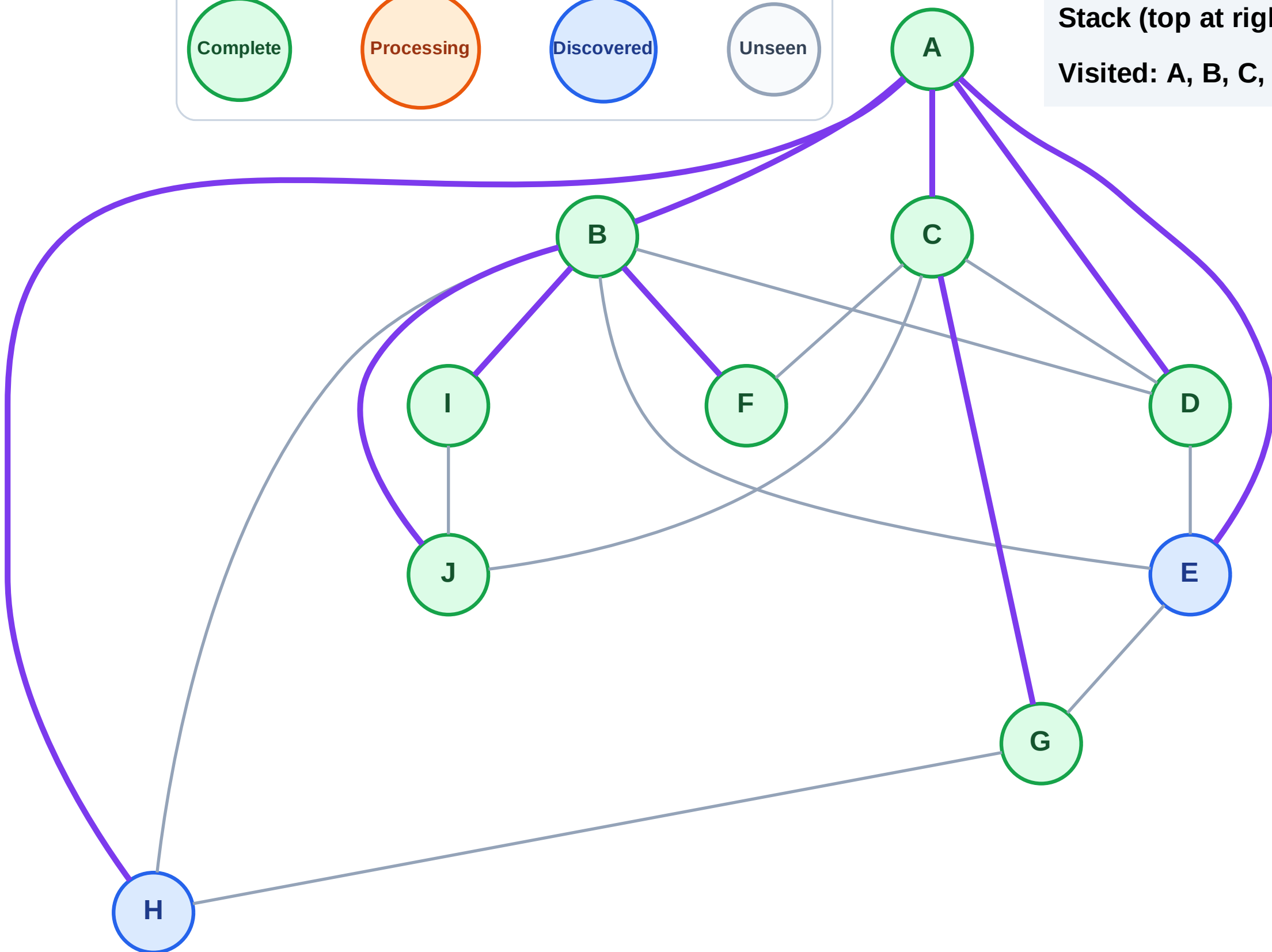
Step 17: No new neighbors from D

Legend



Stack (top at right): H | E

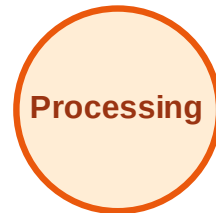
Visited: A, B, C, D, F, G, I, J



DFS Traversal

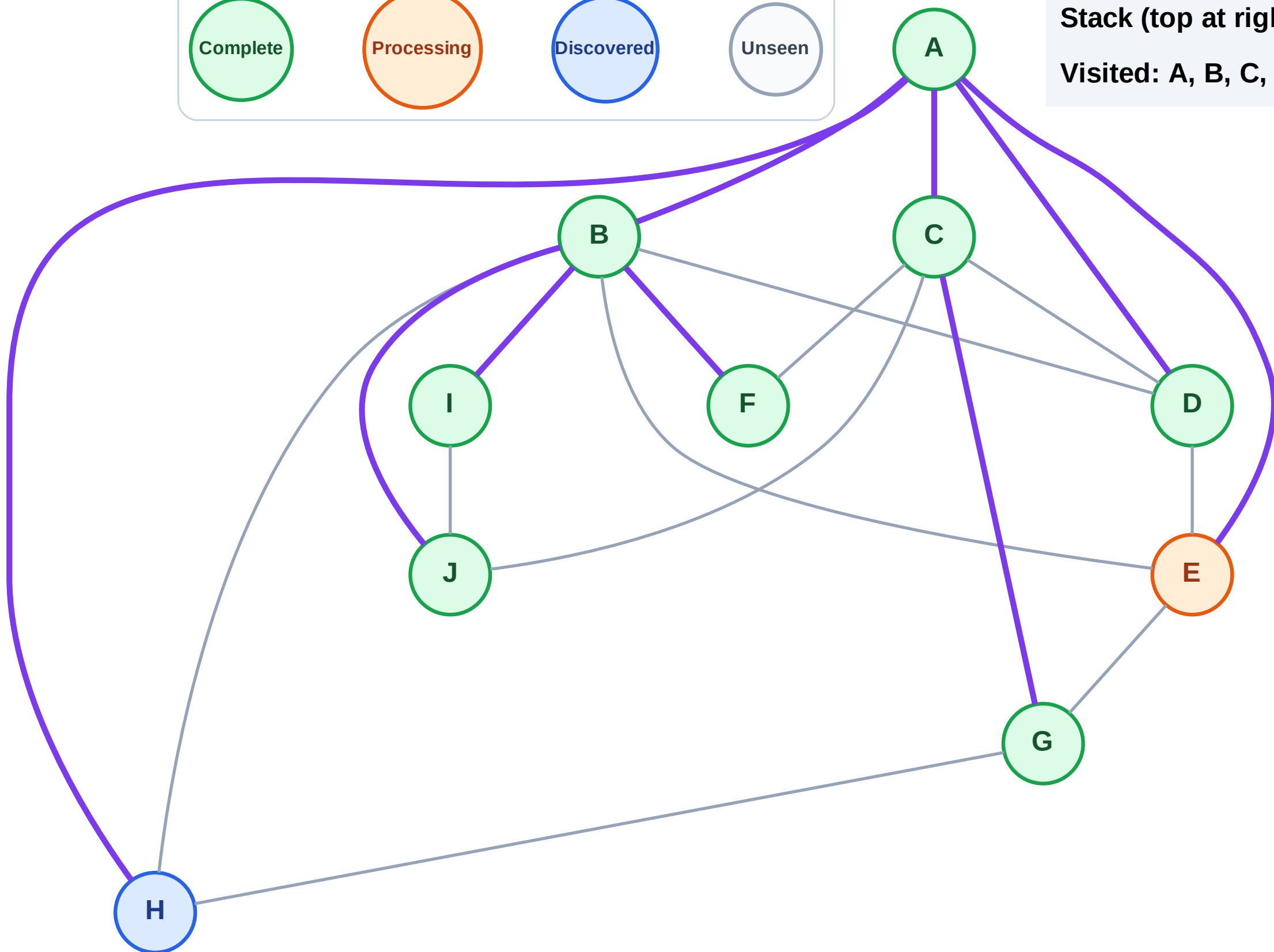
Step 18: Process node E

Legend



Stack (top at right): H

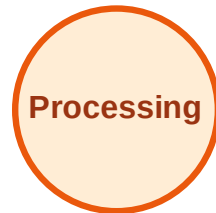
Visited: A, B, C, D, F, G, I, J



DFS Traversal

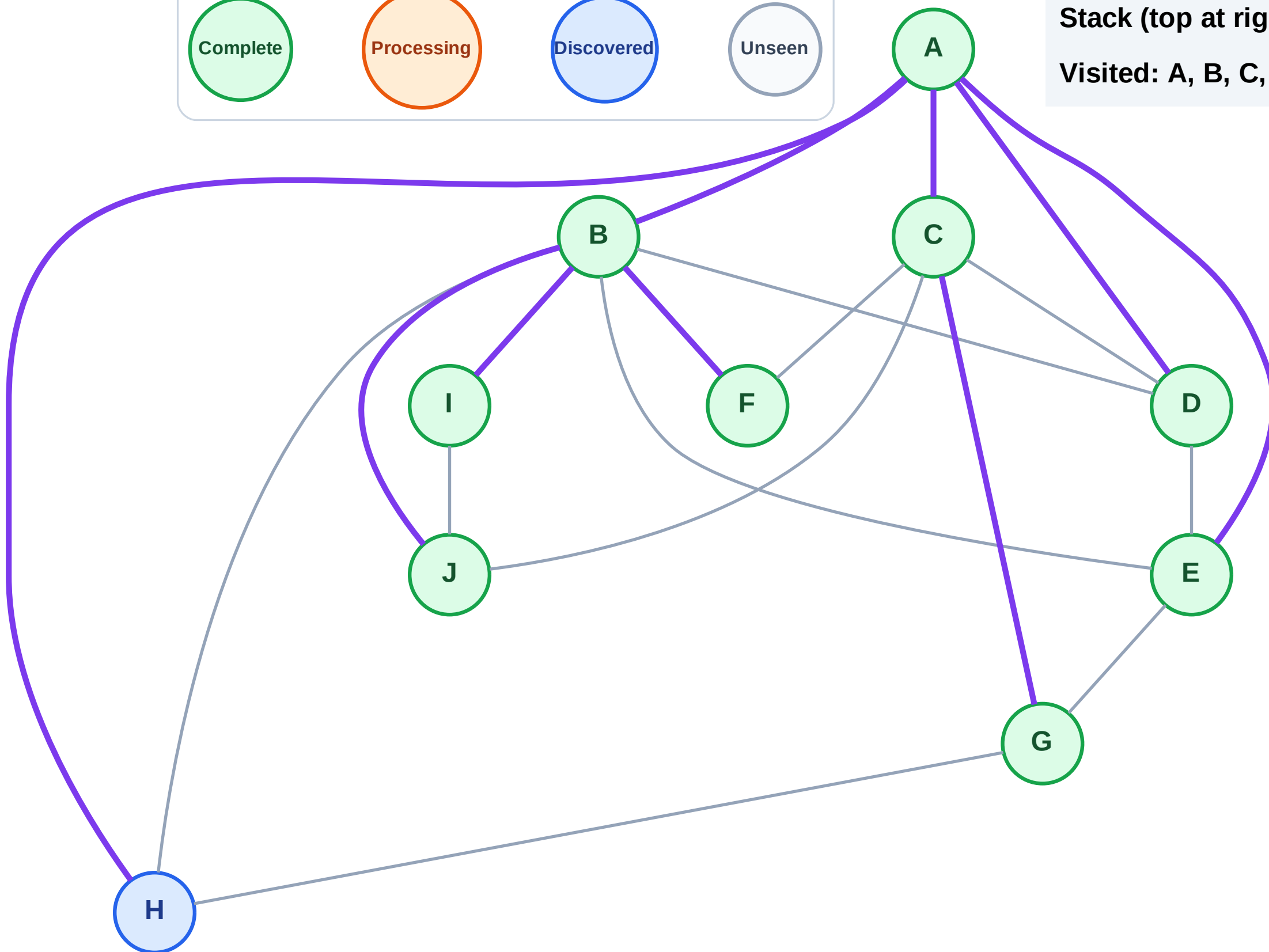
Step 19: No new neighbors from E

Legend



Stack (top at right): H

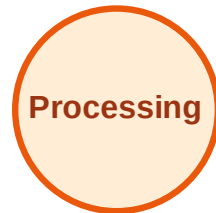
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

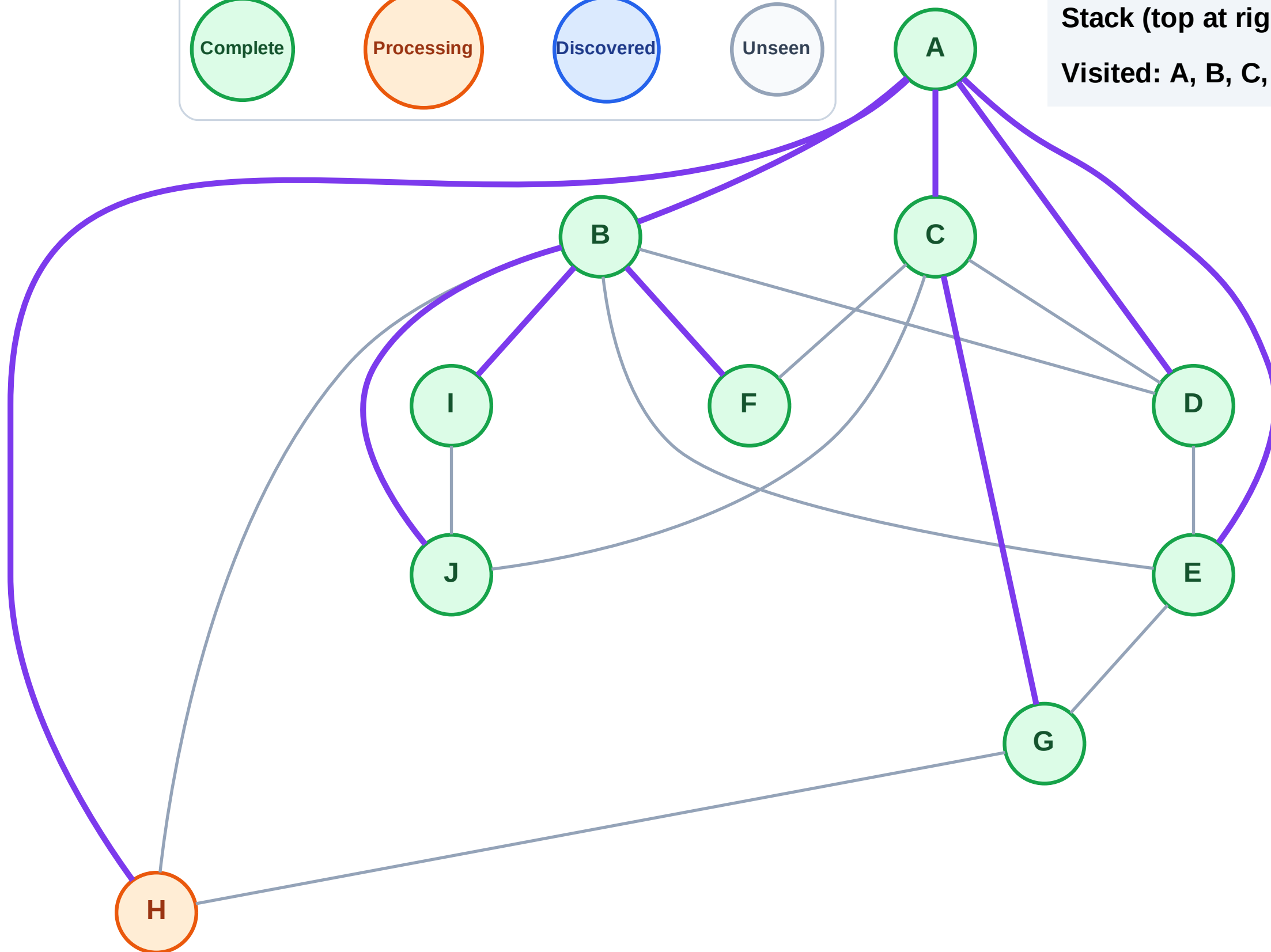
Step 20: Process node H

Legend



Stack (top at right): (empty)

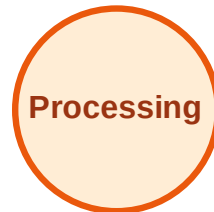
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

